

**Decoding Urban Emissions: A Predictive Analysis of the Influence of
Vehicular Fuel Types on Air Quality Under Varied Meteorological
Conditions.**

Dissertation

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ABSTRACT

This dissertation provides a thorough examination of how vehicular fuel types affect urban air quality under various meteorological conditions. By using exploratory data analysis (EDA) and predictive modeling, the study reveals significant correlations between diesel vehicle emissions, specifically from heavy goods vehicles (HGVs), and higher levels of NO₂ and particulate matter (PM_{2.5} and PM₁₀). The research emphasizes the crucial role of diesel emissions in urban air pollution, with petrol cars also showing a correlation with PM₁₀, but less significantly. Meteorological conditions, such as wind speed, temperature, and humidity, were found to have a significant impact on pollutant dispersion and chemical reaction rates. Granger's causality tests in the study indicate a predictive relationship between diesel HGV emissions and SO₂ levels, indicating the potential of past emissions data in predicting future SO₂ concentrations. The study utilizes advanced machine learning models, particularly Multivariate Long-Short Term Memory Networks (LSTMs), which demonstrate potential in accurately predicting CO and NO₂ levels. However, the study notes challenges in modeling SO₂ and particulate matter, indicating the need for further model refinement. The findings highlight the vital role of weather conditions in regulating pollutant concentrations and the potential of predictive models in managing urban air quality. Although LSTMs show promise, particularly for certain pollutants, the study acknowledges the necessity for continued development to improve accuracy and reliability. This research provides a detailed analysis of the impact of different fuel types on urban air quality, the influence of meteorological conditions, and the advancements in predictive modeling techniques, contributing substantially to the field. It offers valuable insights for policymakers and urban planners to formulate strategies to mitigate air pollution and improve urban air quality.

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Introduction

According to the Transport Energy and Environment Statistics, road vehicles are the single biggest cause of London's air pollution, producing nearly half of all nitrogen oxides and emitting tiny particles of rubber and metal into the air. The choice of vehicle fuel types significantly influences the composition and intensity of emissions, contributing to variations in air quality. From their statistics to select destinations, petrol cars have higher direct emission impact than diesel cars and electric cars mostly on indirect emissions [33]

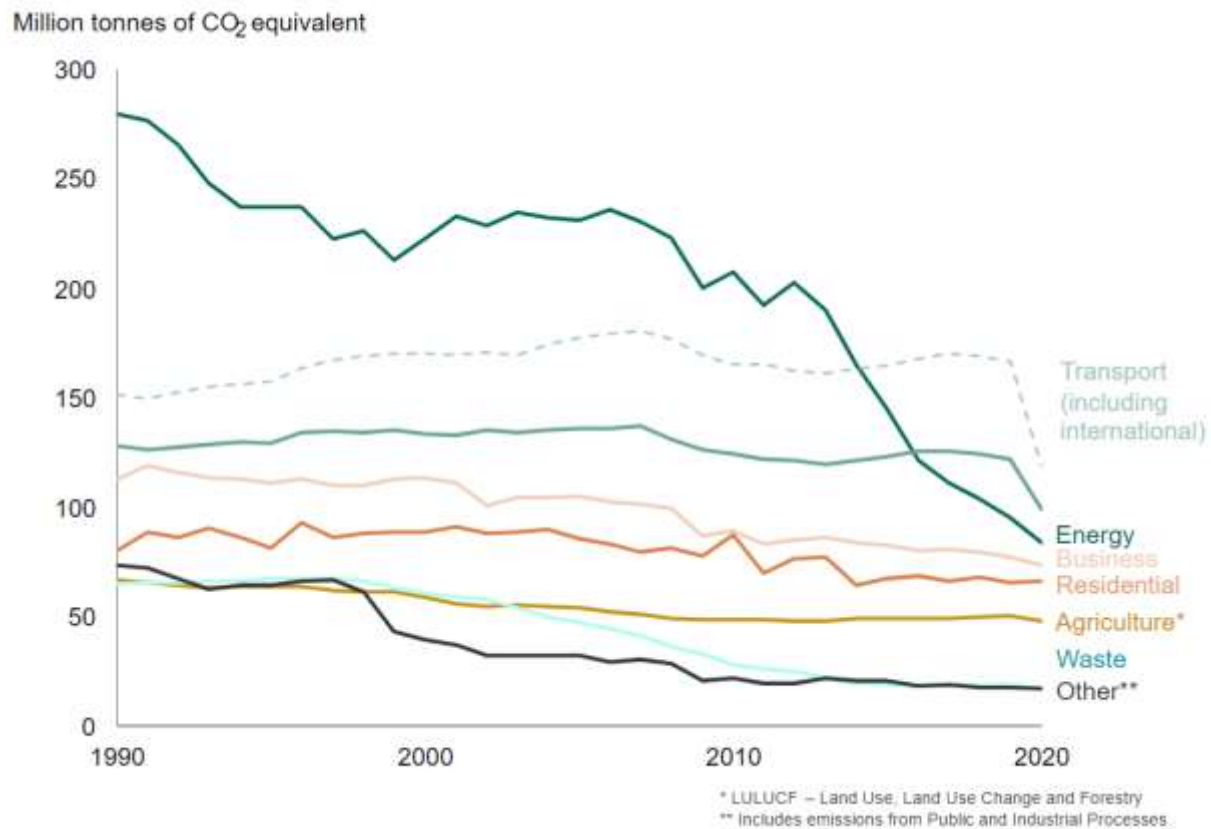


Figure 1 Statistics of GHG emissions from transport. Source: (Transport and environment statistics, 2022, 2023).

With the growing trend of urbanization and increasing reliance on vehicular transportation, the levels of pollutants in the air has increased significantly. This has become a major concern with significant implications for public health, environmental sustainability, and climate change. In densely populated urban areas, the quality of the air we breathe is predominantly influenced by vehicular emissions, a derivative of our growing dependence on various fuel types for

transportation. The combustion of these fuels releases a range of pollutants, each contributing differently to air quality depending on their chemical composition and the prevailing meteorological conditions [37]

Vehicle fuel types exhibit a range of emission characteristics. For example, diesel cars are well known for producing high levels of PM and Nitrogen Oxides (NO_x), two substances that greatly worsen air pollution. However, the main emissions from gasoline-powered automobiles are volatile organic compounds (VOCs) and carbon monoxide (CO) (VOCs). Because they emit fewer hazardous emissions, alternative fuels such as biofuels and compressed natural gas (CNG) are a more environmentally friendly option. According to the UK National Atmospheric Emissions Inventory, NAEI, emissions from road vehicles are determined by a variety of factors, including the fleet's age and makeup, the vehicle's weight or size the emission standards it met when it was new, the technologies used to reduce emissions, the kind and quality of fuel it is powered by, how it is driven, the nature of the trip, and the temperature conditions [27].

Moreover, the effect of these emissions on air quality is not static; it varies under different meteorological conditions. For instance, high temperatures can lead to higher emission rates of some pollutants due to increased fuel evaporation. Additionally, wind speed and direction can affect the way pollutants spread, which can lead to different levels of concentration in the air. For example, when winds are calm and temperatures are low, pollutants can get trapped and worsen the air quality in the local area [26].

The London Borough of Barnet

The London Borough of Barnet presents a special and compelling case for this study. As one of the most populated boroughs in London,, Barnet faces significant challenges associated to urban air pollution. The borough's diverse geography, spanning from crowded urban areas to substantial green spaces, offers a diverse terrain for examining the impact of vehicular emissions on air quality. Barnet's well-developed road infrastructure, which includes significant roads such as the North Circular and M1, there is a substantial amount of traffic congestion, making it a suitable location to investigate the impact of various fuel types on the air quality of urban areas.

Moreover, Barnet's commitment to environmental sustainability, as evidenced by its local air quality management plans and initiatives to reduce carbon emissions, aligns with the objectives of this research. The borough's efforts to monitor and improve air quality provide a rich dataset for analysis. By focusing on Barnet, this study not only addresses a local environmental issue but also contributes to the broader understanding of urban emissions, offering valuable insights for local policymakers and urban planners in Barnet and other metropolitan areas.

This dissertation seeks to unravel the intricacies between emissions from vehicular fuel types and urban air quality, set against the backdrop of varying meteorological conditions. The significance of this research lies in its ability to address the increasing challenges posed by urban air pollution. As cities expand, the number of vehicles on the roads also increases, thus worsening the problem of air pollution [2]. The adverse effects of poor air quality on health are well-documented and can range from respiratory and cardiovascular illnesses to premature death. [3]. Therefore, understanding the relationship between different fuel types used in vehicles and the resultant air pollution is crucial for developing effective strategies to mitigate these health risks.

Additionally, for policymakers, urban planners, environmental scientists, and the general public, it offers a comprehensive understanding of how various types of vehicular fuels contribute to urban air pollution. The study provides valuable insights to develop policies and strategies aimed at improving air quality. Furthermore, the findings can assist urban planners in designing cities that prioritize public health and environmental sustainability.

The approach of this dissertation is twofold. Firstly, it involves a comprehensive analysis of the correlation between different vehicular fuel types and the levels of key pollutants in urban air quality. This analysis is crucial in identifying the major contributors to urban air pollution and understanding their impact. Secondly, the study explores how meteorological conditions modify the impact of emissions from various fuel types on urban air quality. This aspect is particularly important as weather conditions play a significant role in the dispersion and concentration of pollutants in the air.

Furthermore, this research employs predictive modeling to forecast air pollution levels based on the distribution of vehicle fuel types and meteorological data. The employment of advanced machine learning models, such as Multivariate Long-Short Term Memory Networks (LSTMs),

offers a novel approach to predicting air pollution levels using emissions from vehicle fuel type distributions.

In summary, this dissertation provides a comprehensive analysis of the influence of vehicular fuel types on urban air quality under varied meteorological conditions. The findings of this study are expected to contribute significantly to the field of environmental science, offering new insights and tools for combating urban air pollution.

Literature Review

Urban air quality management has undergone significant changes over the years, with emphasis on vehicular emissions due to their major impact on air pollution. The transition of vehicular fuel types from coal-powered vehicles to petrol, diesel, and now electric and hybrid vehicles has marked the history of urban emissions. Each of these transitions has brought about changes in the urban air quality landscape. While earlier efforts focused on reducing visible smog and soot, over time, the focus has shifted to invisible pollutants such as NO_x and particulate matter, which have more serious health effects. [1]

According to the UK government's Transport and Environment Statistics 2022 report, vehicle transportation contributes significantly to emissions of nitrogen oxides, carbon dioxide, particulate matter (PM)₁₀, and PM_{2.5}. Domestic transportation was the source of 99 MtCO₂e (million tonnes of carbon dioxide equivalent) in 2020, the greatest drop in emissions ever recorded and a 19% decrease from 2019 [33].

As a result of improved newer vehicle fuel efficiency, diesel use for road transportation surpassed gasoline use starting in 2005, and the total amount of fuel used for road transportation in the UK stayed mostly stable from 1990 to 2017. During this time, petrol use decreased and diesel use increased emissions of numerous contaminants, including nitrogen, sulphur dioxide, and carbon monoxide, which are especially harmful to health

This is evidence that the type of fuel used by vehicles plays a crucial role in determining the nature and quantity of pollutants emitted. Common pollutants such as Nitrogen Dioxide (NO₂), Carbon Monoxide (CO), Particulate Matter (PM₁₀ and PM_{2.5}), and Sulfur Dioxide (SO₂) vary considerably based on the fuel type, including gasoline, diesel, Compressed Natural Gas (CNG), and newer, cleaner alternatives. Understanding the relationship between different vehicular fuel types and the levels of these key pollutants is essential for effective urban air quality management.

Extensive research has been conducted to evaluate the impact of vehicular emissions on urban air quality, with the focus on the variability in emissions based on vehicle type and fuel. Wu et al. (2022) conducted a study that delves into the complexities of air quality modeling, with a specific focus on the nuances of emissions from various vehicle types. The study highlights how different fuel types (gasoline, diesel, CNG, etc.) contribute to the emission of pollutants like NO₂, CO, PM₁₀, and PM_{2.5}. While the research provides a foundational understanding of the relationship between vehicular emissions and urban air quality, it falls short of offering a detailed analysis of how each fuel type specifically contributes to overall air pollution. Therefore, more granular research is needed to distinguish the impact of each fuel type on urban air quality.

Another research work uses machine learning algorithms to estimate emissions from diesel automobiles, specifically focused on NO_x. In order to minimise the number of models required, the research clusters automobiles with comparable emissions behaviour and develops quick instantaneous models. The study concludes that nonlinear regression and artificial neural networks are equally accurate, and these quick instantaneous models can be applied to high-resolution emissions inventories. The study highlights the potential of these technologies in more comprehensive urban air quality investigations by relying on machine learning models for NO_x prediction. Nevertheless, the study's focus is restricted to diesel cars, indicating the need for a more comprehensive strategy that takes a larger variety of vehicle fuels into account[7].

The major primary air pollutants linked to vehicular emissions include soot (black carbon), carbon monoxide (CO), and nitric oxide (NO), all of which pose significant health concerns [2]. However, recent studies have shown that urban air quality is a highly dynamic and complex issue that is influenced by various factors such as the micrometeorological conditions of the area. The concentration of pollutants like sulphur dioxide (SO₂), nitrogen dioxide (NO₂), and particulate

matter (PM₁₀) can vary significantly across different urban locations [17]making it challenging to address the problem effectively. Furthermore, long-term studies conducted in cities like Seoul have revealed that meteorological conditions, along with emissions, play a vital role in local air quality through accumulation, ventilation, regional transport, and atmospheric chemistry [28].

Furthermore, research has shown that vehicular emissions, in particular, have been found to vary significantly under different traffic conditions and are sensitive to factors like vehicle power, performance, and driving behavior. This means that the same vehicle can produce vastly different levels of pollutants depending on the driving conditions [14]. Additionally, the impact of meteorological conditions on pollutants like ozone (O₃) has been increasingly recognized. Studies have revealed the nonlinear relationships between emissions, atmospheric chemistry, and meteorological factors, highlighting the need for a more comprehensive understanding of air quality dynamics in urban environments [11].

In a recent study, Shakya et al. (2023) integrated meteorological data and vehicle emissions to estimate PM_{2.5} concentrations using deep learning algorithms. The study is novel in that it shows how weather influences pollutant behaviour by including meteorological factors with emissions data. This integration is crucial as weather conditions such as temperature, wind speed, and humidity profoundly affect pollutant behavior. Low temperatures can enhance the formation of secondary pollutants, while high humidity levels can lead to increased particulate matter due to hygroscopic growth. Wind speed and direction are pivotal in pollutant dispersion, often determining the spatial distribution of air pollutants in urban areas. Even said, it raises concerns about the precise ways in which various fuel types interact with different climatic conditions to alter urban air quality, even while it represents a considerable advancement in our understanding of the role of weather in pollutant dispersion.

The use of deep learning models indicates a shift towards more advanced, data-driven approaches in air quality prediction. Khurana et al. (2021) demonstrate the application of deep learning techniques in predicting urban air quality. The research successfully applies these models to urban traffic data, yet it does not fully address the challenge of integrating these predictions with diverse meteorological conditions. This presents an opportunity for future research to explore how these advanced models perform under varying weather scenarios.

Another study suggests various machine learning techniques that generate cost-effective estimations of air quality without depending on pricey sensor observations by combining meteorological and transit-related information. Several machine learning and deep learning models, such as a Random Forest regressor, an Artificial Neural Network with Bayesian regularisation, a linear regressor, and a Long Short Term Memory network, were examined. The studies demonstrated that by utilising meteorological characteristics, such as temperature and humidity, as well as urban-related factors, including vehicle transits, it is possible to provide reasonably priced estimation results over the pollutants. The study showed that most contaminants could be estimated well, and also emphasised the significance of careful preprocessing and data engineering due to the study's challenge of dealing with scattered and missing data.

The reviewed literature collectively lays a solid foundation in understanding the impact of vehicular emissions and meteorological conditions on urban air quality. However, several gaps emerge:

1. **Comprehensive Fuel Type Analysis:** While existing studies address emissions from specific fuel types or vehicles, a comprehensive analysis comparing different fuel types, including their distinct contributions to key pollutants, is lacking.
2. **Detailed Meteorological Impact Study:** The interaction between different meteorological conditions and various fuel type emissions has not been thoroughly explored. There's a need to understand how specific weather patterns influence the dispersion and impact of emissions from different fuel types.
3. **Predictive Model Validation:** While advanced machine learning models show promise, there's a gap in validating these models across diverse urban settings and varying meteorological conditions. This calls for a study that assesses the efficacy of machine learning models in accurately predicting pollution levels.

By addressing these gaps, this thesis adopts a multifaceted approach, integrating correlation analysis, predictive modeling, and extensive data analysis to provide a comprehensive understanding of urban air quality. Unique in its approach. It not only examines the impact of emissions and weather on air quality, but it also looks at the combined effects for all vehicle types with petrol and diesel fuel features. The use of advanced predictive models like Random Forest

Regressor, Support Vector Regressor and Multivariate Long-Short Term Memory Networks (LSTMs) is a significant advancement in the field as they can analyse the complex, non-linear and temporal patterns and interactions that traditional models miss. Additionally, employing metrics such as RSquared, Mean Squared Error (MSE) and Median Absolute Error (MedAE), would enable the study to understand the interpretability performance of the model.

The advantages of this integrated approach are manifold. It allows for a more nuanced understanding of the interplay between emissions from vehicle fuel types and meteorological conditions, acknowledging that the impact of pollutants is not static but varies with changing weather patterns and vehicle fuel features. By employing advanced machine learning techniques, the research aims to overcome the limitations of traditional models, which often fail to capture the complexities of urban air pollution dynamics mostly found in the data. This approach is expected to yield more accurate and reliable predictions, crucial for policymakers and urban planners in devising effective air quality management strategies.

In summary, this thesis represents a significant step forward in urban emissions research. It not only contributes to the existing body of knowledge by filling critical gaps but also introduces innovative methodologies that could redefine how urban air quality is predicted and managed.

The rest of this project paper is organised as follows:

- Section 3, Methodology and Exploratory Data Analysis
- Section 4, Model Development
- Section 5, Result Analysis and Discussion
- Section 6, Conclusion along with The Limitations and Suggestions for Future Work

The final section discusses the conclusion and next projects that might enhance the model's interpretability and performance.

Methodology

The aim of this study is to investigate the complex relationships between air pollution levels in urban areas, the emissions of different vehicle fuel types, and how weather conditions affect air quality. This investigation also extends to assessing the efficacy of various machine learning models in predicting air pollution levels based on a combination of vehicular emissions data and meteorological information. This multidisciplinary approach integrates principles and techniques from environmental science, statistics, and data science to provide a comprehensive understanding of the factors influencing urban air quality.

As seen in figure 2 the study involves data collection, data preprocessing, feature engineering, exploratory data analysis, statistical analysis, modeling and evaluation. The foundation of this study is the set of robust datasets utilised, including the traffic dataset, two air quality datasets, weather data, vehicle fuel type and emissions factors dataset. Each dataset was meticulously gathered from reliable transport and urban environmental monitoring networks and databases to ensure high accuracy and relevance.

The preprocessing and feature engineering phase was thorough and involved cleaning, handling missing and duplicate values, interpolation, extrapolation, creating time lags, and developing a formula to determine the proportion of fuel type for each vehicle type from the hourly counts of traffic data. The EDA stage involved statistical analysis and visualisation to identify patterns and trends among various variables.

The study utilised advanced machine learning modeling techniques like Random Forest Regressor, Support Vector Regressor (SVR), and Multivariate Long Short-Term Memory (LSTM) networks to predict air pollution levels. These models were preprocessed, trained, tuned and evaluated against baselines using metrics like r squared, median absolute error and mean squared error to ensure reliable findings.

It's worth mentioning that Google Collaboratory's Jupyter Notebook, a cloud-based development environment, was used for streamlined code execution and management through the course of the project.

Data collection, preprocessing, and feature engineering are covered in the next section.

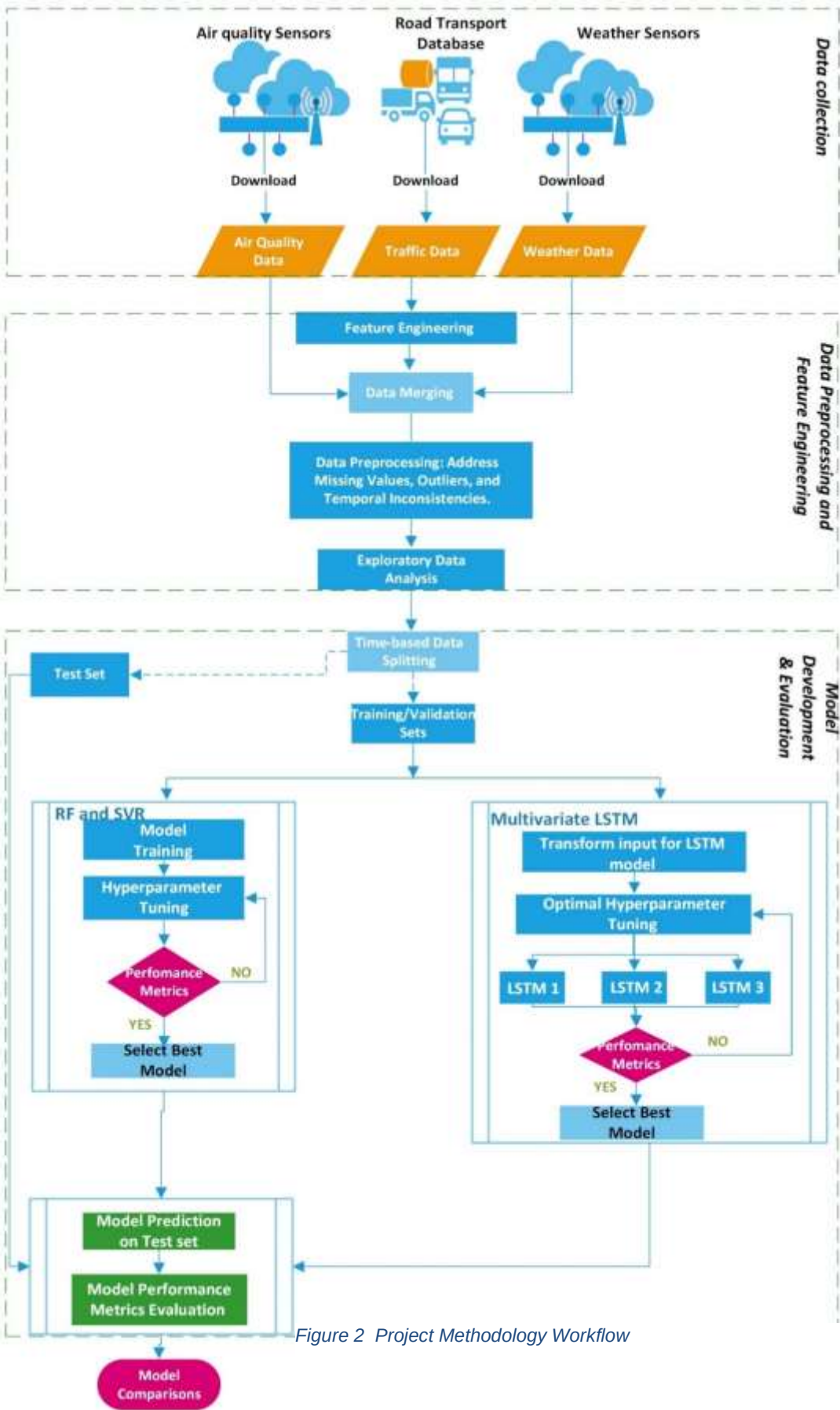


Figure 2 Project Methodology Workflow

3.1 Study Area: The London Borough of Barnet

The study area for this dissertation is the London Borough of Barnet, which happens to be one of the largest of the 32 boroughs in London, with a total area of 86.74 square kilometers. According to the 2021 census, Barnet has a population density of 4,489 people per square kilometer, placing it among the more densely populated outer boroughs of London. [18]

The focus of this study is centered on the two most densely populated parliamentary constituencies in Barnet: Hendon and Finchley & Golders Green. These regions were selected specifically as they experience high traffic volume, which substantially adds to urban emissions and has the potential to affect air quality.

The data for this study was collected from various sensors strategically placed in these densely populated and high-traffic areas as seen in figure 2 .

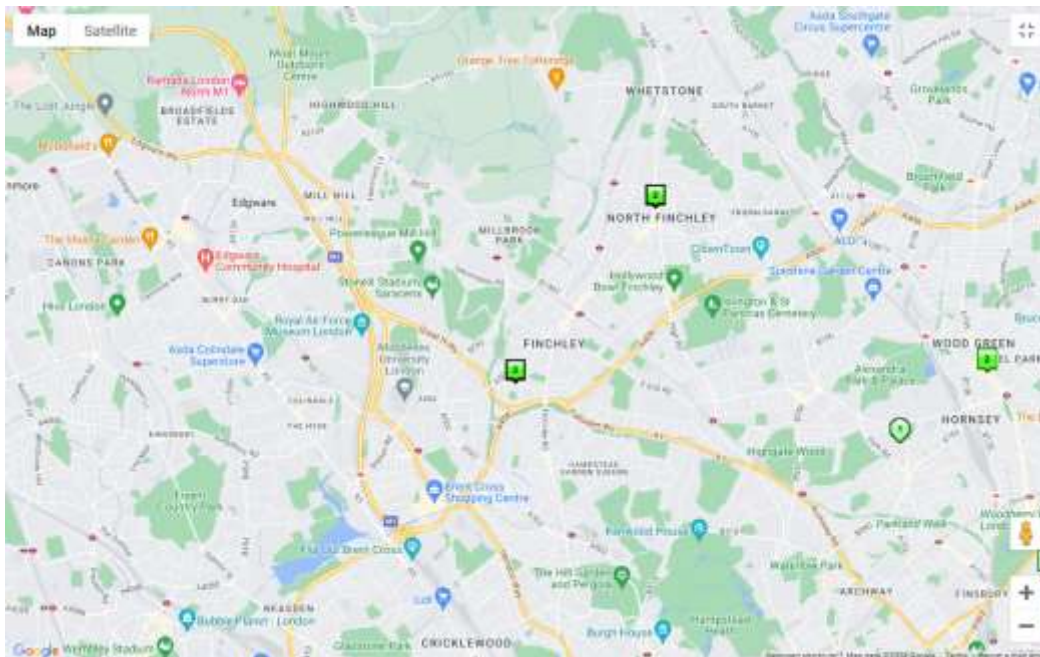


Figure 3 Air quality AURN Sensors in Barnet

3.2 Data

Data quality, volume, and availability are critical factors to take into account when using Machine Learning (ML) to solve problems. To achieve the best results, it was essential to document and review all stages of data collection, processing, feature engineering, merging, visualisation, analysis, preprocessing, and feature selection. Doing so achieved an optimal combination of high-quality datasets and selected features.

3.2.1 Data Acquisition

This study utilises a multifaceted data collection approach, integrating various temporal datasets to comprehensively analyse the interplay between vehicular emissions, air quality, and weather conditions. The datasets are meticulously chosen across all datasets using monitoring sensors within 7 collection sites selected based on the latitude and longitude recorded in the Barnet traffic data. The goal was to ensure consistency in the spatial features and temporal granularity of our model's interpretation. The datasets that have been collected include traffic counts data, air quality data, meteorological data, and for the purpose of feature engineering, emission factors and vehicle licensing statistics data. To conduct this study, the period we utilised spanned months, with hours starting from 07:00 a.m. to 7:00 p.m., in 2021 and 2022. These time periods were selected because they represent peak hours when traffic volume is high, especially after the easing of COVID-19 lockdowns.

Traffic Data: Getting the traffic data was the most challenging task for various reasons most of which has to do with lack of high temporal granularity and absence of the variables required for analysis. However, we finally extracted our dataset from the Department for Transport database. The data is collected using around 8,000 roadside 12-hour manual counts and continuous data from automatic traffic counters, and other automatic traffic counter data sources such as National Highways and Transport for London . The dataset was downloaded in a CSV format and was loaded and stored on our Google Collab python jupyter notebook environment. Although it contained the variables we could work with, the dataset lacked complete temporal granularity and so we had to perform interpolation to fill in missing dates and data points using estimated values that maintain a linear relationship between the neighboring data points. After this process we ended

up complete hourly data with a total of 39324 data points. Descriptive statistics about the dataset can be found in Table 1 of the Appendix

Air Quality Data: The Air quality data was taken from two different sources to cover a comprehensive range of pollutants. These sources include:

- UK's Air Information Resource under the Department of Environment Food and Rural Affairs (DEFRA): The Air Quality England database provides access to the latest air pollutant concentrations from over 1500 monitoring sites in the UK from DEFRA. The monitoring stations in the study area, London Borough of Barnet, are part of the Automatic Urban and Rural Network, which uses various techniques to monitor and validate the readings of air pollutants. The Chalgrove Primary School Monitoring Station was utilised for this study and obtained in a CSV format via an email request. The downside, it only contained two pollutants, NO₂ and PM₁₀ [9].
- OpenWeather: An IoT technology containing data from radars, AQ sensors stations, and satellites. Data was extracted in JSON format using an OpenWeather API and loaded to our workspace using the JSON library on Python. Provides air pollutant measures for NO₂, PM₁₀, PM_{2.5}, SO₂, and CO [23].

Barnet air quality readings were combined with OpenWeather data, resulting in a total of 18192 data points. This integration allowed for a comprehensive analysis of air quality data from various sources and sensors, broadening the scope and providing a better understanding of all pollutants in the study area. The datasets were methodically integrated to ensure consistency and compatibility. Descriptive statistics about the dataset can be found in Table 2 of the Appendix

Weather Data: The weather data was also extracted from OpenWeather in CSV formats. Open weather was sourced from radars, meteorological agencies, weather satellites, and weather stations. The model provides data on essential weather parameters, including temperature, precipitation, wind, etc [24]. Descriptive statistics about the dataset can be found in Table 3 of the Appendix .

Datasets used for feature engineering:

- **Vehicle Licensing Statistics:** The Driver and Vehicle Licensing Agency (DVLA) maintains a vehicle database with details like make, model, fuel type, and colour. The data is collected automatically or manually and updated regularly. The accuracy of recorded features is estimated to be 89% [35]. The VEH0105 data table, which is produced in compliance with professional standards and offers comprehensive quarterly statistics on licenced vehicles by body type, fuel type, and local authority, is the source of the data that we used. It is consistent with the traffic data's vehicle type classifications and provides the fuel type distribution, which is essential for our study.
- **Emission Factors Data:** The emission factors database is maintained by the National Atmospheric Emissions Inventory [6]. As the 2022 emission factors are unavailable, we plan to estimate them through extrapolation using the trends from 2018-2021. While not exact, this approach can offer a reasonable approximation if consistent trends or patterns are present.

3.2.3 Data Preprocessing and Feature Engineering

The most critical stage of modeling is data preprocessing. This stage is essential as it ensures quality and accuracy of the data leading to more reliable and meaningful insights. Due to the large amount of datasets we were working with, our study took a sequential and methodical approach to the cleaning, transforming and creating new features to obtain the variables needed to achieve optimum results after merging. Table 1 shows the actions carried out for each dataset and the final feature space that left us with 15619 data points.

The feature engineering process was carried out with the following assumptions:

1. The estimated distributions of vehicle fuel types are logical and consistent with established patterns or external data sources.
2. Removing columns with a high prevalence of zero counts, such as hybrid electric vehicles and certain vehicle types with specific fuel types, is a logical decision after integrating vehicle fuel type counts. This step ensures the analysis focuses on more prevalent vehicle types and fuel combinations.

3. For emission estimation, since the traffic HGVs counts are combined, we assumed that most of the HGVs in our traffic data are Rigid HGVs, based on the industry standard statistics that 70% of HGVs are Rigid and 30% [19]
4. For emission estimation, we also assumed that most buses are of diesel fuel type [13]
5. For emission estimation, we assumed that most motorcycles are of petrol fuel type [32]

Data	Data Preprocessing	Feature Engineering	Final Feature Space	Model variable type
Traffic Data	- Datetime transformation - Feature selection - dropping irrelevant columns - Handling missing dates - Interpolation - Date filtering - Handling duplicates - Sorting and aggregating - Data type transformation - Renaming columns for clarity - Merge all datasets after feature engineering on the index of traffic data Handling NaNs - filled with zeros	- Quarter assignment to each month - Vehicle Fuel Type-Mapped Vehicle types, Normalized Quaterly proportions from vehicle licensing statistics using this formular: Normalized Proportion = (Average Proportion for the Fuel Type) / (Total Proportion for all Fuel Types) and then applied proportions to traffic data to estimate daily vehicle fuel type proportions. - Vehicle Fuel type Emission Estimation using emission factors and estimated emission	- Date - Diesel_Cars - Petrol_Cars - Diesel_Buses_and_coaches - Petrol_Motorcycles - Diesel_Light_goods_vehicles - Petrol_Light_goods_vehicles - Diesel_Heavy_goods_vehicles - NOx_Diesel_Cars - NOx_Petrol_Cars - NOx_Diesel_Buses - NOx_Petrol_Motorcycles - NOx_Diesel_LGVs - NOx_Petrol_LGVs - NOx_Diesel_HGVs - SO2_Diesel_Cars - SO2_Petrol_Cars - SO2_Diesel_Buses - SO2_Petrol_Motorcycles - SO2_Diesel_LGVs - SO2_Petrol_LGVs - SO2_Diesel_HGVs - CO_Diesel_Cars - CO_Petrol_Cars - CO_Diesel_Buses - CO_Petrol_Motorcycles - CO_Diesel_LGVs - CO_Petrol_LGVs - CO_Diesel_HGVs - PM10_Diesel_Cars - PM10_Petrol_Cars - PM10_Diesel_Buses - PM10_Petrol_Motorcycles - PM10_Diesel_LGVs - PM10_Petrol_LGVs - PM10_Diesel_HGVs - PM2.5_Diesel_Cars - PM2.5_Petrol_Cars - PM2.5_Diesel_Buses - PM2.5_Petrol_Motorcycles - PM2.5_Diesel_LGVs - PM2.5_Petrol_LGVs	Predictors

		factors for 2022 • Time Lags		
Vehicle Licensing Data	• Handling non-numeric markers • Renaming columns for clarity • Data Preparation for estimating vehicle fuel type counts			
Emission Factors Data	• Datetime transformation • Renaming columns for clarity • Time series analysis & 2022 extrapolation - polynomial model was chosen due to non-linear pattern observed during analysis and evaluated using MSE and residual plots when compared to a simple linear model, However it might be slightly overfitting.			
Weather Data	• Datetime transformation • Feature selection - dropping irrelevant columns • Handling missing values • Date filtering • Handling duplicates - removal • Data type transformation Renaming columns for clarity	Time_lags	• temperature • pressure • humidity • wind_speed • wind_dir • rain_vol	Predictors
Air Quality Data	• Datetime transformation • Feature selection - dropping irrelevant		• CO • SO2 • PM2.5 • NO2	Targets

	- columns - Date filtering - Sorting and aggregating - Data type transformation - Renaming columns for clarity - Merging Air Quality Datasets-Alignment and concatenation:		PM10	
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Table 1 Data Preprocessing and Feature Engineering

Table 2 below is the Feature Selected Set we used for modelling after performing Random Forest Feature Importance.

S/N	RF Feature Selected Set	Description
1	NOx_Diesel_Cars	Nitrogen oxides emissions from diesel cars.
2	NOx_Diesel_HGVs	Nitrogen oxides emissions from heavy goods vehicles running on diesel in g/km
3.	SO2_Diesel_Cars	Sulfur dioxide emissions from diesel cars in g/km
4	SO2_Petrol_Cars	Sulfur dioxide emissions from petrol cars in g/km
5	SO2_Diesel_HGVs	Sulfur dioxide emissions from diesel heavy goods vehicles in g/km
6	CO_Diesel_Cars	Carbon monoxide emissions from diesel cars in g/km
7	CO_Petrol_Cars	Carbon monoxide emissions from petrol cars in g/km
8	CO_Diesel_HGVs	Carbon monoxide emissions from diesel heavy goods vehicles in g/km
9	PM10_Diesel_Cars	Particulate matter (PM10) emissions from diesel cars in g/km
10	PM10_Petrol_Cars	Particulate matter (PM10) emissions from petrol cars in g/km
11	PM10_Diesel_HGVs	Particulate matter (PM10) emissions from diesel heavy goods vehicles in g/km
12	PM2_5_Diesel_Cars	Fine particulate matter (PM2.5) emissions from diesel cars in g/km
13	PM2_5_Petrol_Cars	Fine particulate matter (PM2.5) emissions from petrol cars in g/km
14	PM2.5 Diesel HGVs	Fine particulate matter (PM2.5) emissions from diesel heavy goods vehicles in g/km
15	Temperature	Ambient temperature in degree celsius
16	Pressure	Atmospheric pressure in hPa
17	Humidity	Ambient humidity in %
18	Wind_Speed	Speed of the wind in mm
19	Wind_Dir	Direction of the wind in degree

Table 1 Model Feature-Selected Set

3.3 Exploratory Data Analysis and Model Selection:

Exploratory data analysis (EDA) is pivotal for comprehending the data, recognizing patterns, choosing features, confirming model assumptions, and shaping predictive model designs. During this stage, we scrutinize key variable distributions, explore correlations, observe temporal trends, and detect outliers or uncommon patterns in the data to gain valuable insights, which are essential for building effective models and ensuring their reliability and accuracy..

Distributions:

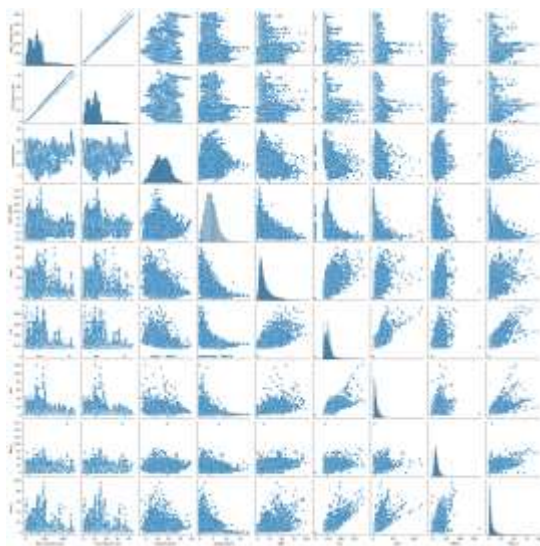


Figure 4 Pairwise plot of some variables emissions

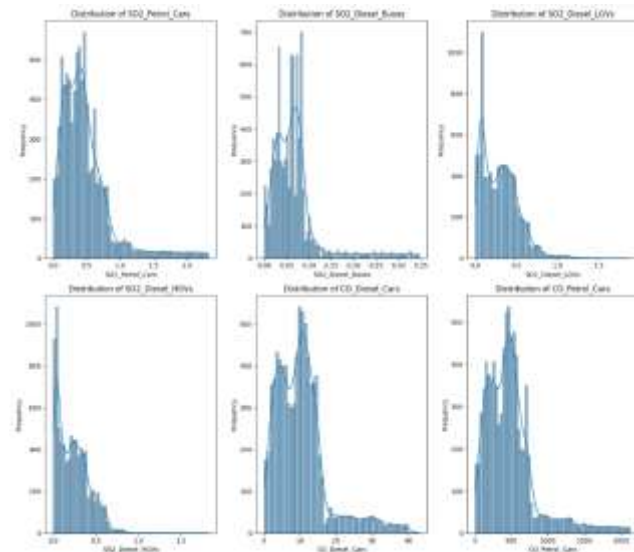


Figure 5 Histogram chart of vehicle emissions

In the pairwise visualization in figure 4 linear patterns in scatter plots suggest a linear correlation between the variables, while more dispersed points suggest a weaker or no correlation. The only variables that seem to be normally distributed are the weather variables, as evidenced by the bell-shaped histograms, whereas others are positively skewed. A deeper dive into each variable with the histogram plots, see figure 5, revealed that the vehicle emission and air pollutants appear right-skewed. This skewness in the data might be influenced by outliers or extreme values. The Shape of the histograms also suggests that most of the vehicles emit a relatively small amount of pollutants, but some may be responsible for higher emissions.

Anomaly Detection:

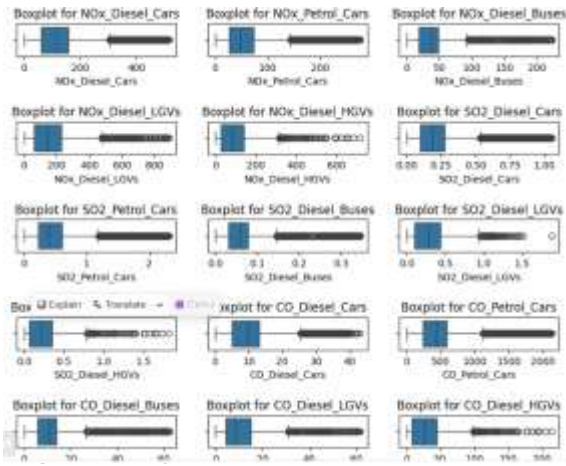


Figure 6

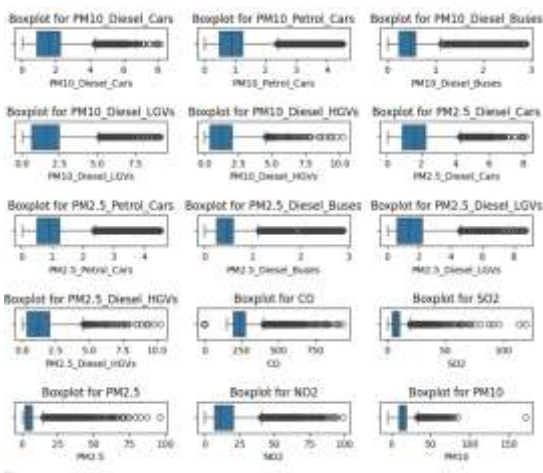


Figure 7

The boxplots in figure 6 & 7 show a right-skewed distribution, with varying median values and a long right tail with outliers, indicating occasional high emissions from the various vehicle fuel types. These outliers may point to extreme but important events or conditions such as traffic peak hours that lead to elevated pollution levels.

Time-series Analysis:

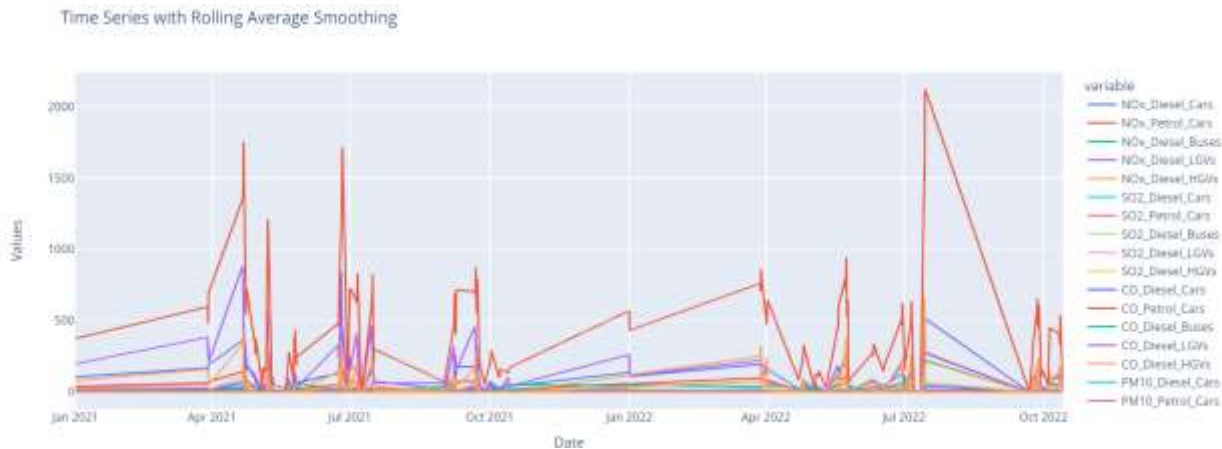


Figure 7 Time Series Plot of vehicle emissions Over The Years

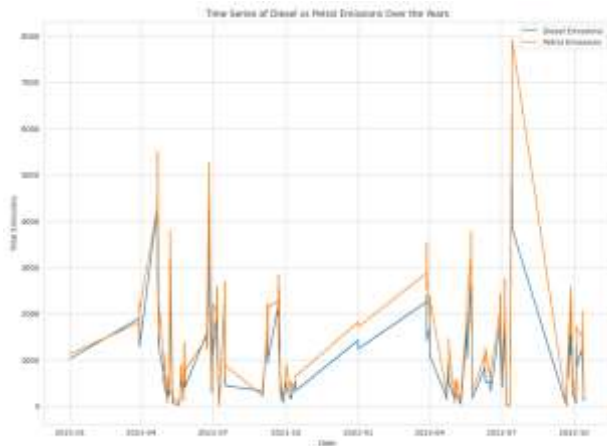


Figure 8 Time Series Plot of the most prevalent fuel type

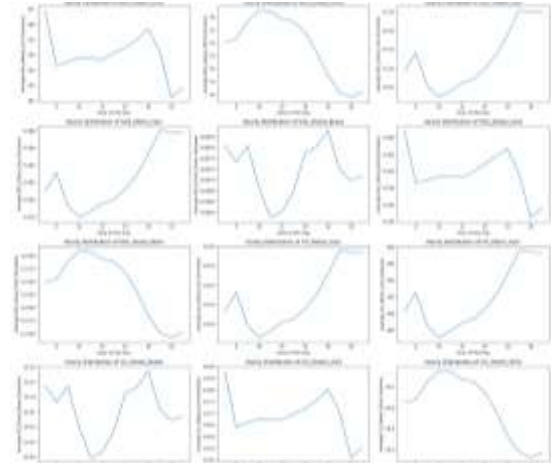


Figure 9 Hourly distribution of vehicle emissions

The time series plots of figure 7 and figure 8 identify visible trends over time, and from the visualisation we understand how vehicle emissions and air quality indicators evolve. The patterns observed shows similar spikes especially from the months of april to july in both years. This could mean that those months were period of holidays which will then lead to increased traffic counts hence the spike in the emissions. Figure 8 suggests that the most emissions are come from petrol vehicles. The line chart of hourly vehicle emissions in figure 9 indicates similar pattern in all emissions from same vehicle types during peak hours. This suggests that emissions are indeed influenced by the time of day, which could be due to various factors such as traffic flow changes during rush hours, different driving conditions, or even variations in atmospheric conditions affecting pollutant dispersion.

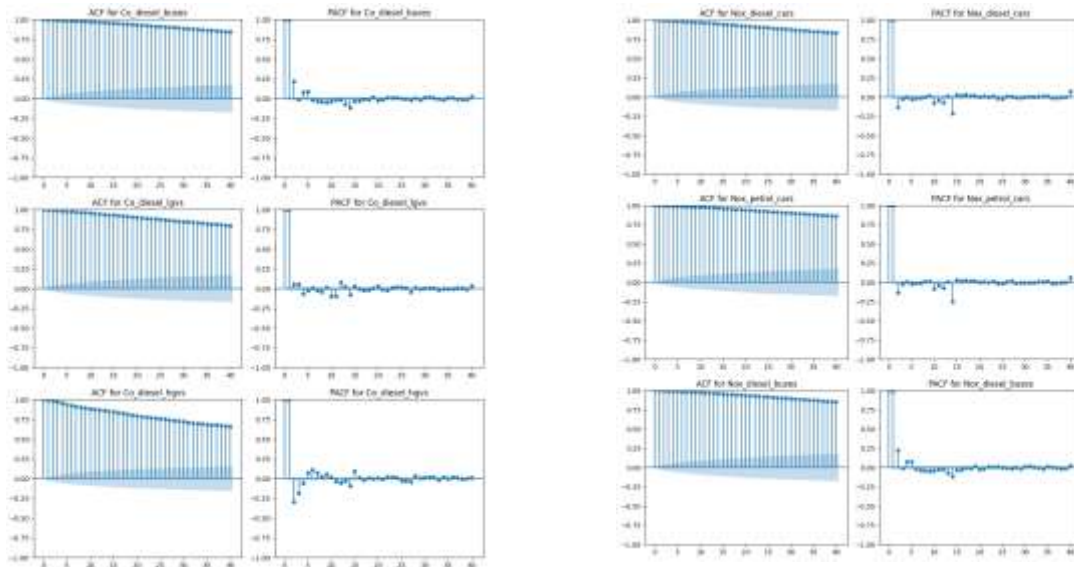


Figure 10 ACF and PACF showing correlation between temporal features

The Autocorrelation Function (ACF) and Partial autocorrelation Function (PACF) plots in figure 10 are commonly used in time series analysis to identify patterns and relationships within the timeseries data, understanding the temporal patterns, identifying the presence of seasonality, and selecting appropriate models for forecasting or analyzing time-dependent data.

The ACF plot shows the correlation between a time series and its lagged values(the values of a variable at previous time points), k , while the PACF plot shows the direct relationship between

$$ACF(k) = \frac{\sum_{t=1}^{N-k} (Y_t - \bar{Y})(Y_{t+k} - \bar{Y})}{\sum_{t=1}^N (Y_t - \bar{Y})^2}$$

an observation and its lagged values, controlling for other lags.

Where Y_t is the value at time t , $\bar{Y}$ is the mean of the series and N is the number of observations.

In Figure 10 most of the Autocorrelation Function and Partial Autocorrelation function plots for all the Vehicle emissions and Air Pollutants are similar. This suggests that for the ACF plots, the pattern from the lags suggests that there is a strong positive autocorrelation at regular intervals (peaks), which could imply a seasonal pattern in the data and also that past values have a long-

term influence on future values. The PACF plot shows a significant spike at lag 1, which drops off sharply and becomes insignificant(although the confidence interval is almost invisible). This indicates that there is a short-term and immediate correlation between a value and its immediate predecessor. It also suggests that machine learning models, such as autoregressive models or recurrent neural networks, may be suitable for capturing the immediate correlations between air pollutant levels, vehicle emissions, and weather data.

Correlation Analysis:

As part of our EDA we have employed Spearman's rank correlation coefficient to investigate the relationship between the variable fuel types and air pollution levels. Spearman's correlation is a non-parametric measure of rank correlation that assumes how well the relationship between two

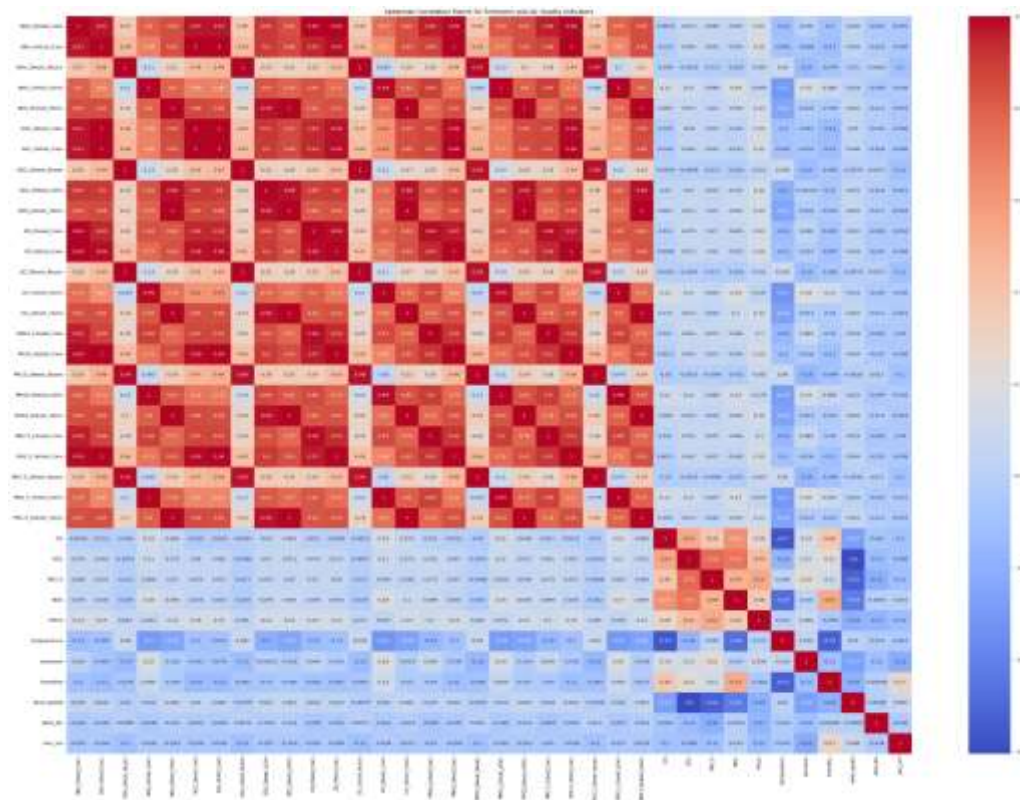


Figure 11 Correlation Plot for Vehicle Emissions + Air Pollutants + Weather Conditions

variables can be described using a monotonic function.

Given that our data did not conform to normal distribution the Spearman correlation coefficient was an appropriate choice as it does not require normality in data. The correlation coefficients, ρ are between -1 and +1, with +1 indicating a perfect positive rank correlation, -1 being a perfect negative rank correlation, and 0 indicating no correlation.

In the Spearman Correlation Matrix we've observed the following:

- **High Positive Correlation between Different Pollutants from the Same Vehicle Fuel Types:** Our findings demonstrate a high positive correlation among various among various pollutants emitted by the same fuel type. For instance, NO_x emissions from petrol cars show a strong positive relationship with their CO and PM_{2.5} emissions. Scientifically, this could be indicative of a shared variance due to common emission factors such as the combustion efficiency of engines, the quality of fuel being used or similarities in driving styles and conditions, all of which are pivotal in the emission generation process.
- **Moderate Positive to Low Negative Correlation between Emissions and Air Quality Indicators:** Not overlooking the Positive relationships between these variables, the low and negative correlations between vehicle emissions and air quality indicators are quite weak (ranging from -0.009 to -0.01 for different emission types), indicating a less direct or more complex relationship between these specific emissions and ambient pollutant levels. This could be due to atmospheric reactions, dispersion factors, or the presence of other pollutant sources.
- **Strong Positive Correlation between Air Quality Indicators:** SO₂ shows a strong positive correlation with PM_{2.5} (0.760787) and NO₂ (0.738563), suggesting that sources contributing to SO₂ emissions are also contributing to these other pollutants. This may indicate common sources such as industrial activities or the burning of fossil fuels.
- **Moderate to Low negative Correlation between Vehicle Emissions and Weather Data:** Positive correlations (close to 1) between vehicle emissions and weather variables suggest that certain meteorological conditions are associated with higher emissions. For example, temperature has a positive correlation with NO_x from diesel cars, this means that warmer temperatures lead to increased NO_x emissions. Negative correlations (close to -1) suggest that certain emissions decrease as the weather variable increases. For instance, a negative correlation between humidity and NO_x emissions might indicate that higher humidity

levels are associated with lower NO_x emissions, possibly due to atmospheric chemical reactions that reduce NO_x under humid conditions.

Statistical Analysis:

Statistical analysis quantifies the strength and relevance of the correlations between the data, verifying first impressions from visual analysis and validating or invalidating the patterns seen visually. This allows for a more rigorous evaluation of the relationships between the data. We performed two rigorous tests to support our findings from the visual analysis using:

Multiple Regression Analysis: To quantify the correlation between multiple independent variables (various vehicle fuel types emissions, weather conditions) and a dependent variable (key pollutant levels).

Carbon Monoxide (CO)

The model accounts for 32.6% of CO variability, with diesel vehicles significantly increasing CO levels. A one-unit increase in CO from diesel cars leads to a substantial 20.6196 increase in CO levels. Interestingly,

weather variables like temperature and wind speed, despite having high p-values, show a negative correlation, indicating they help reduce CO levels.

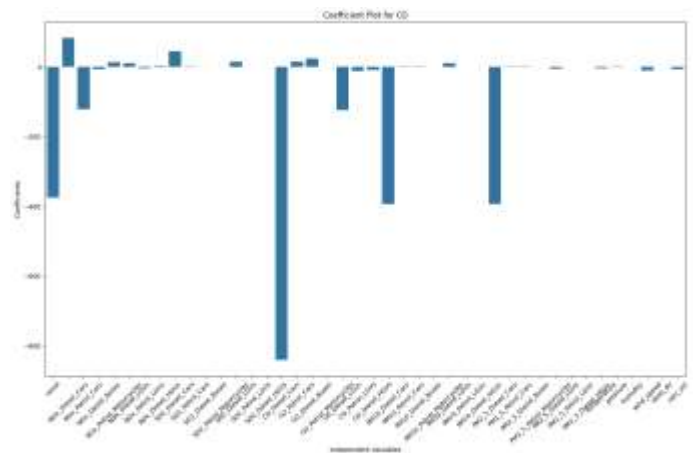


Figure 12 OLS Coefficient plot for CO

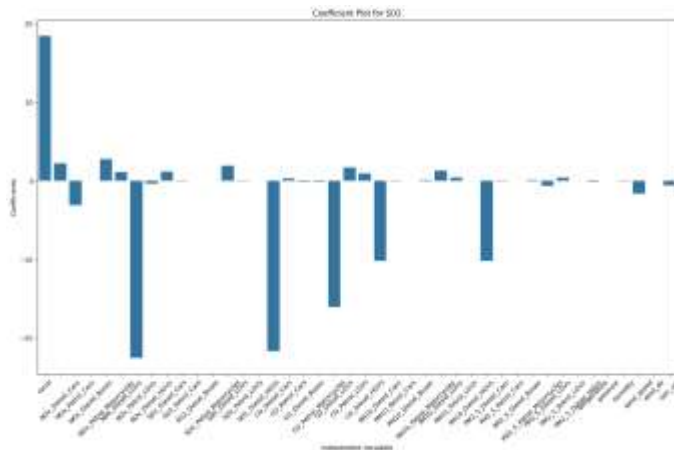


Figure 13 OLS Coefficient plot for SO2

Nitrogen Dioxide (NO2)

The model better explains NO2 variability with an R-squared of 0.348. Diesel vehicles, especially NOx emissions from diesel light goods vehicles (HGVs), are major contributors, increasing NO2 levels by 7.0207 per unit.

Weather conditions like wind speed and temperature inversely affect NO2, while humidity and wind direction have a direct impact.

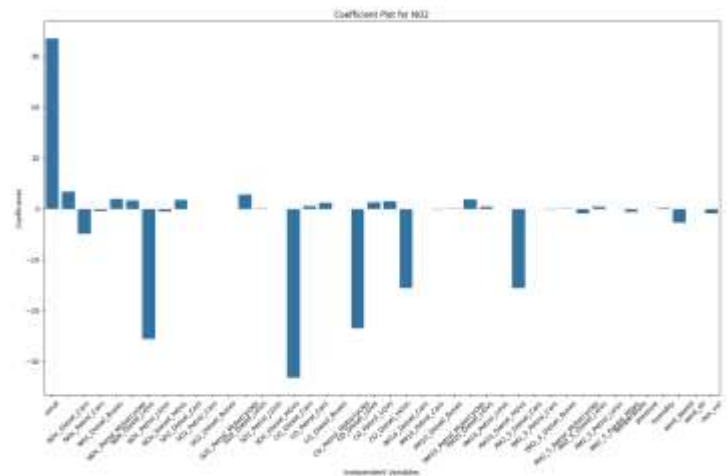


Figure 14 OLS Coefficient plot for NO2

Particulate Matter <10 microns (PM10)

The model explains 15% of PM10 variance, with diesel vehicle emissions being significant predictors, highlighting their role in PM10 levels.

Weather variables also play a significant role, indicating their influence on PM10 concentration.

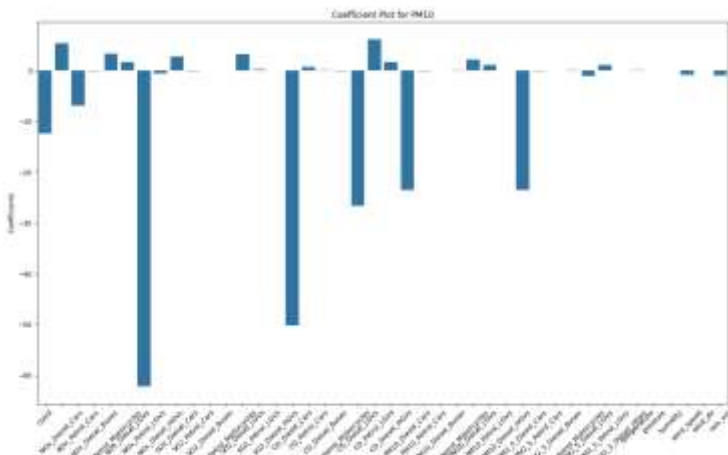


Figure 14 OLS Coefficient plot for PM10

Particulate Matter <2.5 microns (PM2.5)

The model has an R-squared of 0.217, indicating a moderate explanation of PM2.5 variability. Mostly Diesel car emissions are key contributors.

Weather conditions, including wind speed, direction, temperature, and rain volume, are significant predictors, affecting PM2.5 levels.

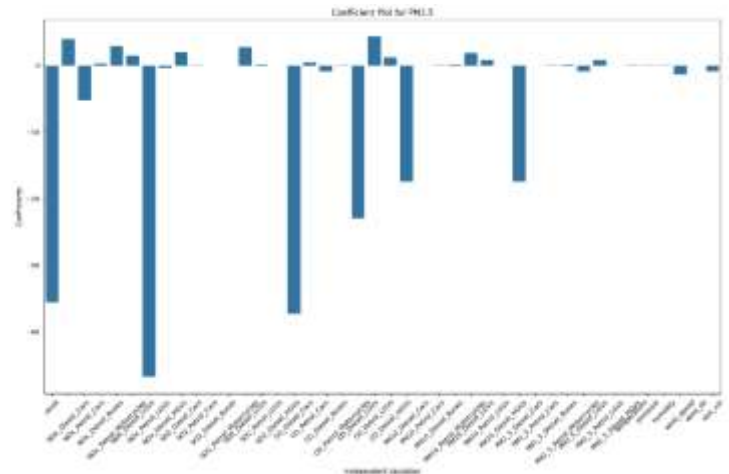


Figure 15 OLS Coefficient plot for PM2.5

The regression analysis results support the results that we deduced from the correlation analysis. However, the analyses are limited by multicollinearity problems; additionally, the residuals' high skewness and kurtosis suggest that the data are not normally distributed, indicating that more sophisticated statistical testing is required to validate our conclusions. To statistically investigate the relationships further, we will conduct a "Granger's causality test.

Granger's Causality Test: This test would help us understand, through inference, if there is and the strength of a relationship between our time series variables

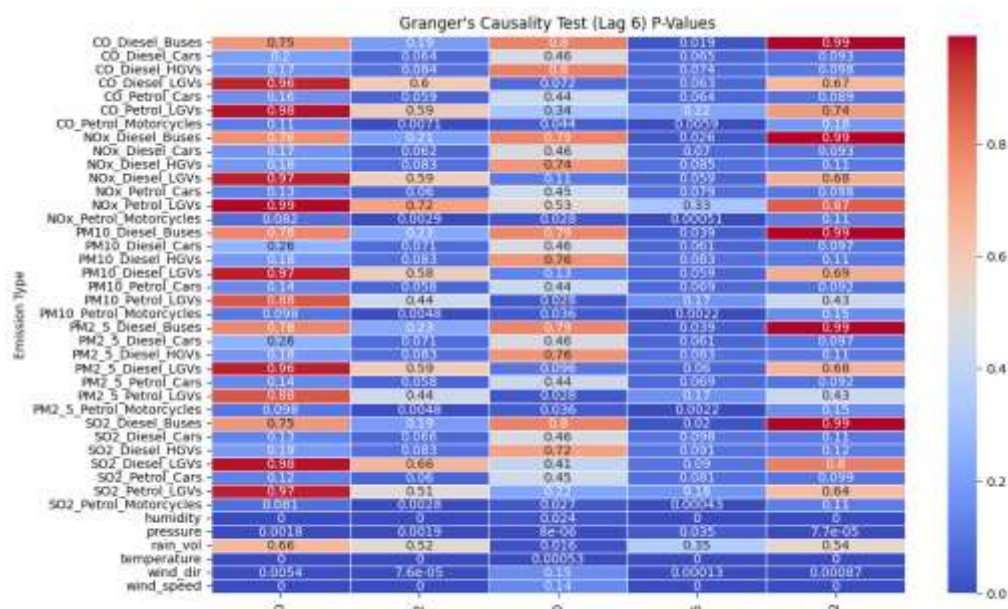


Figure 16 Granger's Causality test at lag 6

The Granger causality test up to lag 6 reveals significant relationships between vehicle fuel type emissions, meteorological conditions, and air pollutant levels. Notably, emissions from diesel cars and buses show a strong influence on NO₂, PM₁₀, and PM_{2.5} levels, indicating a direct impact on air quality. Meteorological factors like temperature, wind speed/direction and humidity also exhibit varying degrees of influence, with temperature showing a notable effect on all pollutant levels with extremely high P-values. These findings underscore the intertwined nature of vehicle emissions and weather conditions in determining air pollutant concentrations.

3.4 Model Selection and Development

3.4.1 Algorithm Selection Rationale: The data we're dealing with is complex, so we needed algorithms that can handle non-linearity, large feature spaces, and time-series data. We chose three algorithms: Random Forest, Support Vector Regression (SVR), and Long Short-Term Memory (LSTM) networks. We chose Random Forest because it's robust against overfitting and is effective in handling large feature spaces and multicollinearity. We chose SVR because it excels in dealing with non-linear data through the use of kernel functions. We chose LSTM networks because they are ideal for time-series data and are capable of capturing the long-term dependencies and patterns. These algorithms complement each other and offer a comprehensive approach to predicting vehicle fuel type emissions under varied weather conditions.

3.4.2 Model development:

Random Forest: When handling complex data, ensemble learning models builds a set of models and merge them together to achieve a better performance using the right ensemble strategies [21]. For our project we used the Random Forest algorithm by Breiman (2001), one of the most popular ensemble learning methods that operate by creating a multitude of decision trees (n-estimators) at training time, with each tree built from a bootstrap sample of the train data, thereby introducing randomness into the model [4].

The algorithm's random selection of data points and features for splitting at each node of the decision trees is a key feature that reduces the model's variance, making it less susceptible to overfitting.

The final prediction is obtained by simultaneously feeding the test data into each tree and then aggregating their predictions by averaging them [21]. This harnesses the collective power of multiple models. This technique effectively captures complex relationships between variables, which is crucial for handling large datasets with higher dimensionality, as demonstrated by Cutler et al. in 2007.

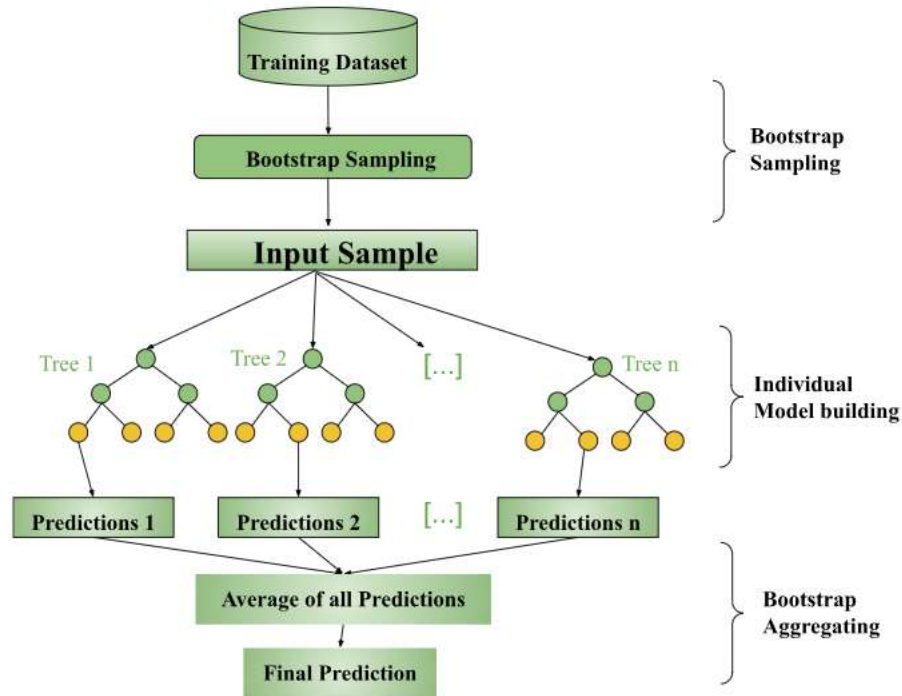


Figure 17 Random Forest Model Architecture

For our model, 60 **n_estimators** were used to determine the number of trees in the forest. Although a higher number of trees could improve the performance, its likely to overfit, so we sought a balance to avoid overfitting while maintaining predictive power through hyperparameter tuning.

The Maximum Depth of Trees of each individual tree in the forest was controlled by setting the **max_depth** to 30. This was done to prevent overfitting and improve the generalisation of the model.

Support Vector Regressor: Support Vector Regression (SVR) is an extension of Support Vector Machine (SVM) used for regression tasks. It's rooted in statistical learning theory and structural

risk minimization principles, hence its objective to minimize the generalisation error and not just the training error observed, making it robust to overfitting especially in high-dimensional spaces [34], [12].

SVR uses a kernel function to map input data to a high-dimensional feature space and finds a linear hyperplane that best fits the data. The approach handles non-linear relationships by linearly modeling them in a higher-dimensional space. As seen in figure 18 SVR uses the ε -insensitive loss function to manage noise in the training data and can address multicollinearity by penalising model complexity. It is well-suited for datasets with multicollinearity and non-normal distribution [31]; [5].

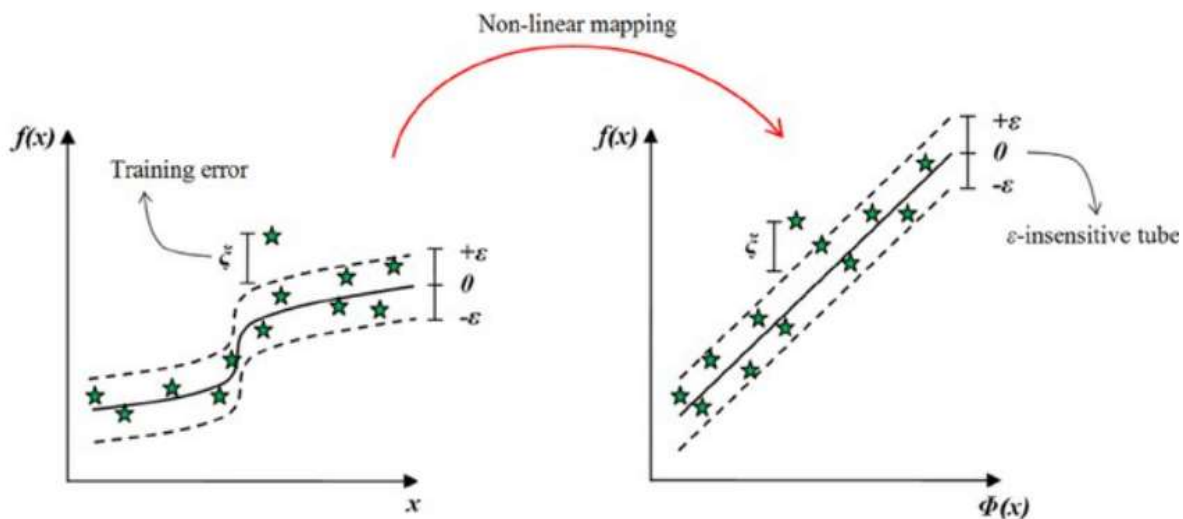


Figure Non-Linear Support Vector Regression. Source: [22]

The Support Vector Regression (SVR) Hyperparameters were critically picked after hyperparameter tuning. The ‘rbf’ kernel was utilized based on the non-linear nature of the dataset and used to transform the input data into a higher dimensional space suitable to improve the model’s performance.

The strength of the **regularization parameter, C**, was set to 0.1 on train and after tuning, controlling the trade-off between maximising the margin and minimising the error. A smaller C creates a smoother decision boundary (less overfitting), while a larger C aims to classify all training examples correctly, potentially leading to overfitting.

Multivariate Long-Short Term Memory Networks: Multivariate long-short term memory networks are a type of recurrent neural networks (RNNs) that are highly suited for complex time-series forecasting tasks because they are especially good at modelling sequential data with multiple input variables [16].

The architecture of a multivariate LSTM for time series modeling consists of four layers: Input, LSTM, Dense, and Output. The Input layer processes the input data sequence-wise, where each sequence comprises multiple time steps, and each time step contains multiple variables (features for each datetime).

The LSTM layer serves as the processing center of the model, containing units with a cell state and three gates (input, output, and forget gates) that regulate the flow of information. The LSTM units also learn the temporal dynamics from the sequences, as well as the interactions between different variables at each time step, which is crucial in scenarios where the output is influenced by a combination of factors.

Following the learning phase in the LSTM layer, the output is passed to the Dense layer, which, in our case, consists of five dense layers for prediction. The weights of these layers are adjusted to learn non-linear combinations of the features learned in the LSTM layer. Finally, the design of the Output layer is based on the prediction requirement.

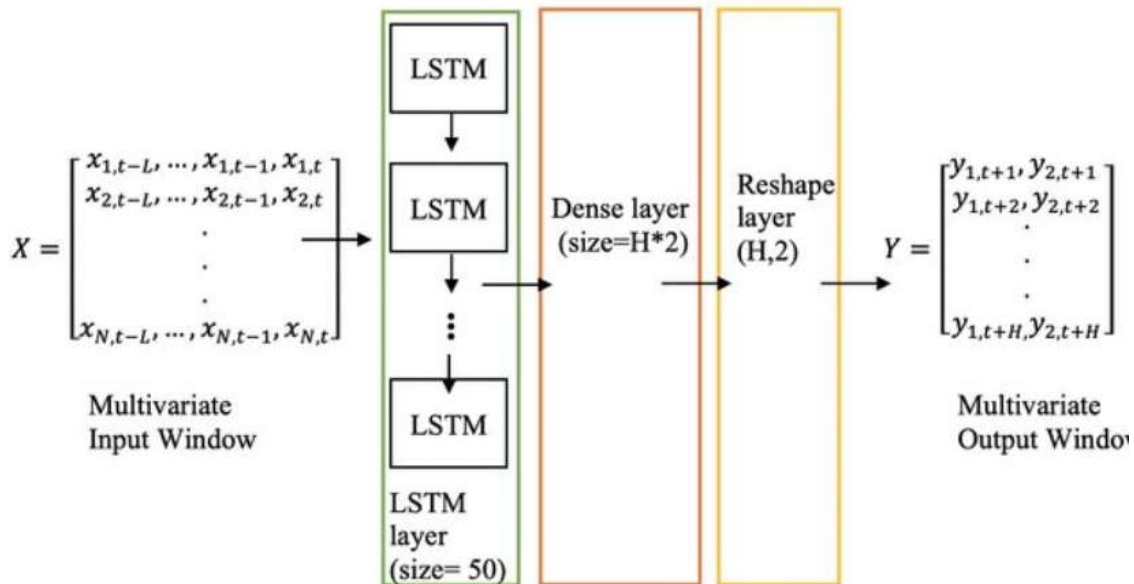


Figure 19 A Multivariate LSTM Time series Model Architecture with Time Step [25].

The final model was implemented with 2 LSTM layers, although more layers can increase the model's capacity to learn complex and intricate relationships, the the risk of overfitting would be increased in our model .

The **neurons** in the LSTM layer represent the units. After tuning, these units are set to 50, which determines the model's ability to learn and identify patterns in the data. Higher number of units can enhance the model's representational power, but it also increases the chances of overfitting.

Dropout Rates are used to prevent overfitting in neural networks by randomly setting a fraction of input units to 0 at each update during training. We utilized a dropout rate of 0.4 for the final model to find the right balance between preventing and retaining enough information for the model to learn effectively.

The **Adam Optimizer** is a highly regarded optimisation algorithm that is widely used for computing adaptive learning rates for each parameter. It was utilised in our model and was preferred over the 'adamax' algorithm during tuning. The Adam Optimizer is known for its fast convergence and excellent performance, making it a popular choice among researchers and practitioners alike.

The **Learning Rate** is a critical hyperparameter that determines the step size taken at each iteration while moving toward the minimum of the loss function. It essentially controls the speed at which the machine learning model learns and updates its parameters. The tuning process involves exploring a range of values between $1e-4$ and $1e-2$, and finally settling on a value of $1e2$. A smaller learning rate can ensure more precise convergence, but it may come at the cost of slower learning and training times. Therefore, finding an ideal learning rate often requires a balance between speed and accuracy.

Batch normalisation is a technique to improve the training speed and stability of neural networks. It helps normalise the activations of the previous layers.

```
Best Hyperparameters:
Number of LSTM layers: 1
Number of neurons in each LSTM layer: 20
Dropout rate: 0.4
Optimizer: adam
Learning rate: 0.006461395303882178
Final Model Architecture:
Model: "sequential_1"
```

Layer (type)	Output Shape	Param #
lstm_2 (LSTM)	(None, 1, 20)	5360
dropout_2 (Dropout)	(None, 1, 20)	0
batch_normalization (Batch Normalization)	(None, 1, 20)	80
lstm_3 (LSTM)	(None, 100)	48400
dropout_3 (Dropout)	(None, 100)	0
batch_normalization_1 (Batch Normalization)	(None, 100)	400
dense_1 (Dense)	(None, 5)	505

```
=====
Total params: 54745 (213.85 KB)
Trainable params: 54505 (212.91 KB)
Non-trainable params: 240 (960.00 Byte)
```

Figure 20 Architecture of the Multivariate LSTM Model

Finally we built a sequential neural network that comprises of two LSTM layers. It is followed by dropout and batch normalization layers and is developed for a regression task with five possible outputs. The model has a total of 54,745 parameters, out of which 54,505 are trainable, and 240 are non-trainable.

3.4.3 Model Training

Random Forest, Support Vector Regression (SVR), and Long Short-Term Memory (LSTM), were trained on preprocessed time-based split datasets. To evaluate feature importances, a preliminary Random Forest model was trained. In order to guarantee that all models concentrate on significant predictors, the most pertinent features were chosen using a threshold set at the median importance level. Each model is trained and evaluated using metrics like R2, Mean Squared Error (MSE), and Median Absolute Error (MedAE). For hyperparameter tuning, a grid search is performed over a range of values, optimizing for the negative median absolute error in the case of Random Forest and SVR. For LSTM, a hyperband tuner is used to explore various configurations of the number of neurons, dropout rates, number of layers, and optimizer settings. The best model is selected

based on the validation loss, ensuring that the model captures the temporal dynamics in the data effectively while generalising well to new data.

3.4.4 Model Prediction using the Test Sets

As part of the model testing phase, the accuracy and generalization ability of the Random Forest (RF), Support Vector Regression (SVR), and Long Short-Term Memory (LSTM) models were evaluated on both tuning and validation and test datasets respectively. To assess the predictive capability of the RF and SVR models, different metrics such as Mean Squared Error (MSE), Median Absolute Error (MedAE), and R2 score were used on the tuning and validation sets. These metrics provided insights into the models' accuracy and ability to generalize to unseen data. The RF model demonstrated its effectiveness in capturing underlying patterns in the data with specific R2, MSE, and MedAE values for both tuning and validation sets. Similarly, the SVR model's performance was evaluated to reveal its strengths and weaknesses in forecasting the target variables. In the case of the LSTM model, the test set was reshaped to match the LSTM's input requirements, and predictions were made. The LSTM's performance on the test set was evaluated by calculating the loss on the Mean Squared Error (MSE) between the model's predictions and the actual target values.

Performance Metrics:

Mean Squared Error (MSE): MSE measures the average of the squares of the errors, i.e., the average squared difference between the predicted values and the true values. Lower values indicate better model performance.

$$\text{MSE} = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

Where y_i is the true value and $\hat{y}$ is the predicted value and is n total number of data points.

Median Absolute Error (MedAE): MedAE represents the median of all absolute differences between the predicted and actual values. It is less sensitive to outliers in comparison to MSE. The calculation of the

median absolute error over n samples is as follows, assuming that y_i represents the matching true value and $\hat{y}$ represents the $\hat{y}_n$ th sample's predicted value.

$$\text{MedAE} = \text{median}(|y_1 - \hat{y}_1|, |y_2 - \hat{y}_2|, \dots, |y_n - \hat{y}_n|)$$

R2 Score: This metric indicates the proportion of variance in the target variables that the model is able to explain. Ranges from $-\infty$ to 1. Higher values are better, with 1 indicating a perfect fit.

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2}$$

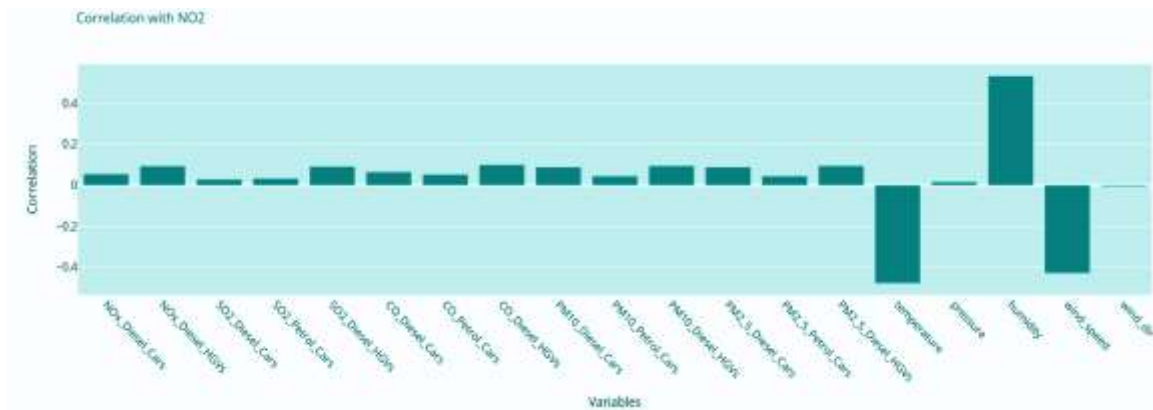
Where $\bar{y}$ is the average of the average of the true values.

Result Analysis & Discussion

In this section, we analyze the results of our study so far for individual air pollutant indicators. By examining each pollutant separately, we aim to uncover insights into their varying levels and the factors influencing them based on the selected features. This approach allows for a detailed understanding of each pollutant's behavior and interactions under varied weather conditions, providing a comprehensive view of air quality dynamics.

The charts below explore the correlations between the air pollutants and the feature-selected set

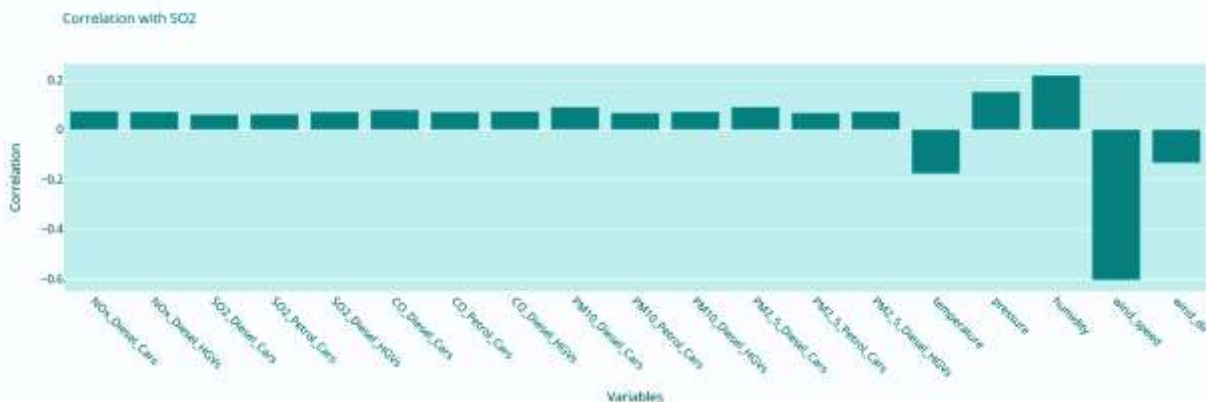
NO₂:



Diesel Vehicles Show a stronger positive correlation with NO₂, particularly from diesel cars and HGVs. This suggests diesel fuel combustion as a significant source of NO₂. There is also a strong correlation observed between NO₂ and weather variables in the plot. A positive correlation suggests high humidity in the atmosphere would lead to higher levels of NO₂ and the negative correlations with wind speed and temperature suggests an inverse relationship where higher readings lead to lower levels of NO₂ in the air.

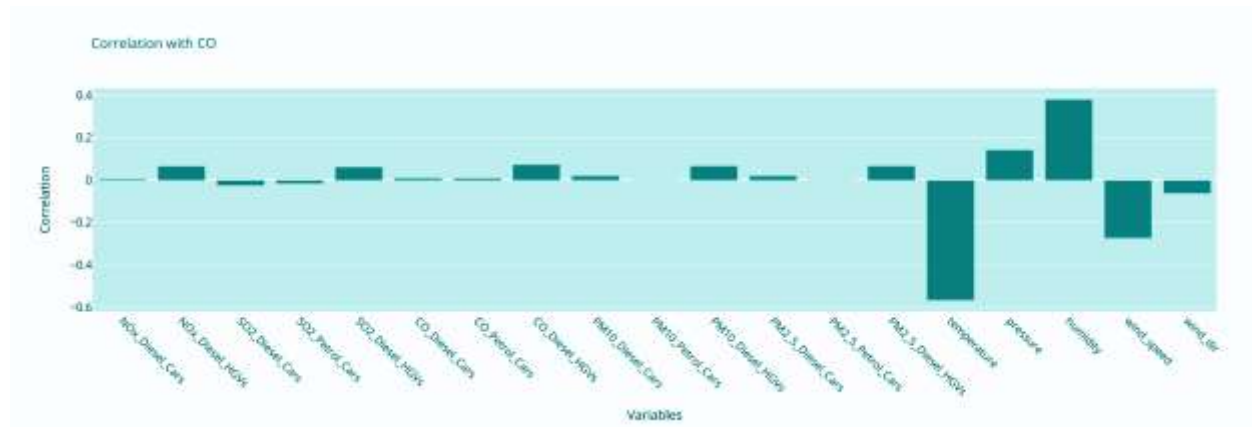
SO₂(Sulphur

Dioxide):



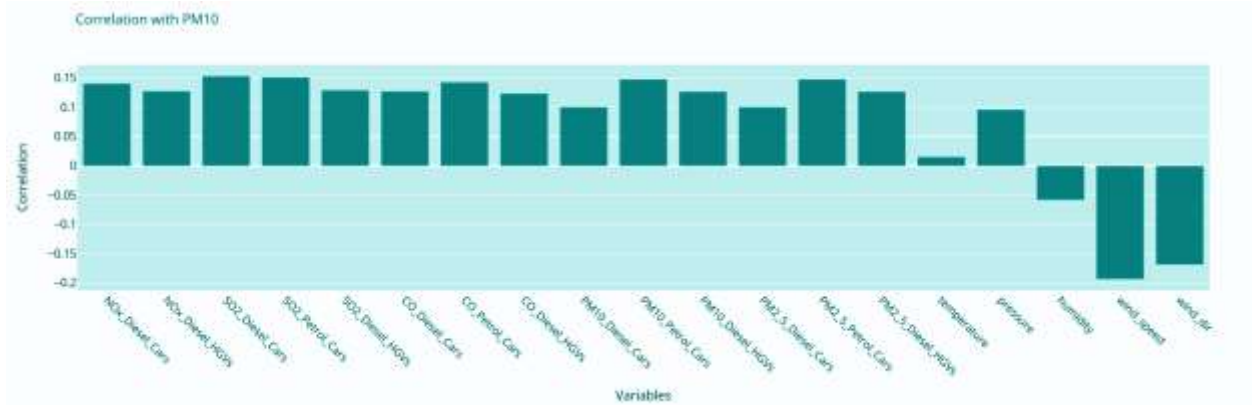
Extremely low correlations across the board for vehicle emissions suggest that vehicular fuel types may not be a primary source of SO₂ in urban environments and that a strong negative correlation with wind speed indicates that increased wind speed significantly reduces SO₂ concentrations, likely through dispersion.

CO(Carbon Monoxide):



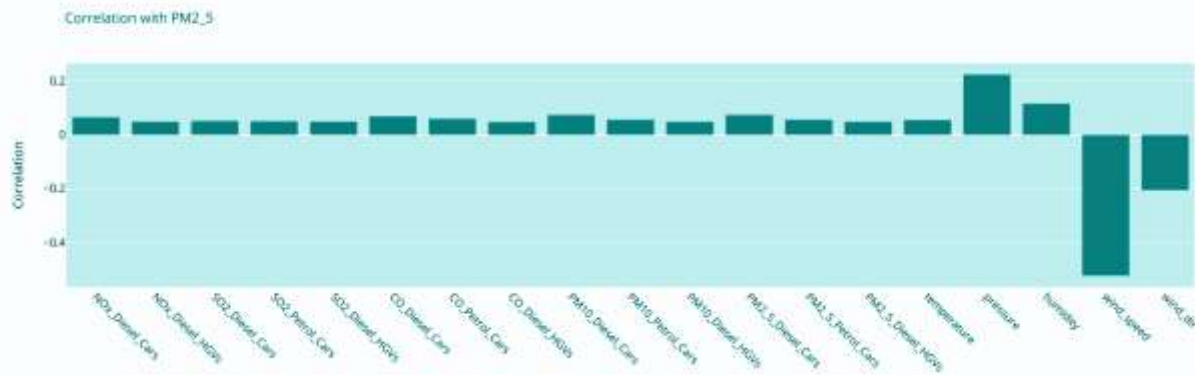
The correlations with vehicle types are generally weak, implying a less direct impact of fuel types on CO levels compared to other pollutants. Strong negative correlation with wind speed, indicating that wind plays a significant role in dispersing CO emissions, irrespective of the fuel type.

PM10(Particulate Matter $\leq$ 10 micrometers):



Petrol Cars show a slightly higher positive correlation compared to diesel, indicating they might be a noteworthy source of PM10. Negative correlations with wind speed and a slight negative trend with humidity suggest that PM10 levels decrease with higher wind speeds and humidity, which facilitate the dispersion and possibly the deposition of particles

PM2.5(Particulate Matter $\leq$ 2.5 micrometers):



There is a strong positive correlation with PM2.5 from diesel cars, indicating that diesel fuel combustion is a significant contributor to fine particulate matter in urban air. Positive correlations seen with temperature could indicate that higher temperatures might increase PM2.5 levels, possibly due to enhanced atmospheric reactions leading to secondary particulate formation. There is also a significant negative correlation with wind speed, suggesting that like PM10, PM2.5 concentrations are also reduced by increased wind speeds.

Figure 21 is a heatmap of p-values from a Granger's Causality Test, showing the statistical significance of the causality between different types of emissions (from various fuel types) and weather variables on the concentrations of key pollutants (CO, NO2, PM10, PM2.5, and SO2). In Granger causality tests, a low p-value (commonly below 0.05) indicates that the row variable Granger-causes the column variable, i.e., it has a predictive value for the latter within the context of the model used.

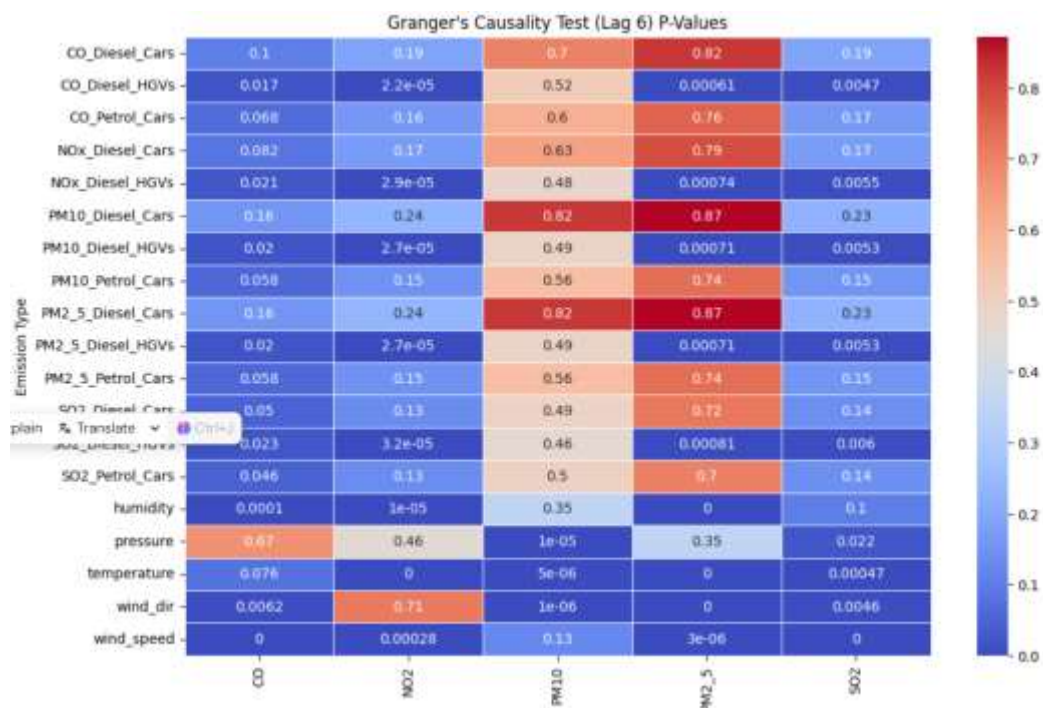


Figure 21 Granger's Causality on Modelling Features

CO (Carbon Monoxide): Very low p-values suggest that emissions from diesel HGVs have a significant predictive value for CO levels, indicating causality also wind speed has a significant p-value, indicating it is a predictor of CO levels.

NO₂ (Nitrogen Dioxide): Emissions from both diesel cars and HGVs show low p-values, indicating they are significant predictors of NO₂ levels , moreover wind direction and wind speed have low p-values, suggesting these weather conditions can significantly predict NO₂ levels.

PM₁₀ (Particulate Matter ≤ 10 micrometers): Emissions from diesel HGVs show the lowest p-values, indicating a significant predictive relationship with PM₁₀ levels whereas pressure and wind speed have significant p-values, suggesting they are important factors in predicting PM₁₀ levels.

PM_{2.5} (Particulate Matter ≤ 2.5 micrometers): Similar to PM₁₀, diesel HGV emissions show low p-values with PM_{2.5}, indicating a significant predictive relationship while temperature and wind speed show very low p-values, suggesting they have a significant influence on PM_{2.5} levels.

SO₂ (Sulfur Dioxide): Emissions from diesel vehicles show low p-values, indicating they are significant predictors of SO₂ levels whereas humidity and wind direction have low p-values, suggesting they are significant predictors of SO₂ levels.

From the EDA, The correlation plots and Granger's causality test together suggest that diesel vehicle emissions, particularly from heavy goods vehicles (HGVs), have a significant predictive relationship with the levels of key pollutants like NO₂, PM₁₀, and PM_{2.5} in urban air quality. Weather variables, notably wind speed and temperature, also play a crucial role in influencing pollutant levels, Next we'll analyze the results of our model predictions to see what insights we could derive.

NO₂:

Model	R ²	MSE	MedAE
Random Forest Regressor	0.064	0.586	0.597
Support Vector Regressor	0.405	0.372	0.377
Multivariate Long-Short Term Networks	0.26	0.021	0.09

Table 2 NO₂ Performance Evaluation

The Random Forest Regressor resulted in limited predictive power, falling short of the project's goal to accurately forecast NO₂ levels, with an R² of only 0.064. On the other hand, the Support Vector Regressor showed a significant improvement, capturing 40.5% of the variance (R² = 0.405) with moderate errors (MSE = 0.372, MedAE = 0.377). This aligns better with the project's predictive aspirations. Lastly, the Multivariate Long-Short Term Memory Networks exhibit the lowest error rates (MSE = 0.021, MedAE = 0.09), indicating a high level of precision in the predicted NO₂ values, although not as strong in variance explanation with an R² of 0.26.

SO₂:

Model	R ²	MSE	MedAE
Random Forest Regressor	0.064	0.586	0.597
Support Vector Regressor	0.230	0.781	0.489
Multivariate Long-Short Term Networks	0.710512	0.066854	0.192066

Table 3 SO₂ Performance Evaluation

The Random Forest Regressor and Multivariate Long-Short Term Memory Networks show limited capability with low R² values, failing to meet the project's requirements for accurate predictions. The Support Vector Regressor achieves a better R², but the errors suggest it deviates more from actual SO₂ values compared to its performance with NO₂.

CO:

Model	R ²	MSE	MedAE
Random Forest Regressor	0.064	0.586	0.597
Support Vector Regressor	-4.265	1.245	0.887
Multivariate Long-Short Term Networks	0.198	0.032	0.095

Table 4 CO Performance Evaluation

The Random Forest Regressor has an R² value of 0.064, indicating a poor understanding of the variability in CO levels, which is not ideal for precise predictions. The Support Vector Regressor has an R² of -4.265, which is worse than a naive mean-based estimator. Its high MSE of 1.245 and MedAE of 0.887 indicate significant prediction errors, which are not helpful for the project's objectives. On the other hand, the Multivariate Long-Short Term Memory Networks show promising results with an R² of 0.198, indicating that the model can account for about 20% of the variance in CO levels. With a relatively low MSE of 0.032 and MedAE of 0.095, the LSTM model suggests a high degree of predictive accuracy, aligning well with the project's goal of accurately modeling the impact of vehicular emissions and meteorological factors on urban air quality.

PM10:

Model	R ²	MSE	MedAE
Random Forest Regressor	0.064	0.586	0.597
Support Vector Regressor	-0.020	1.051	0.629
Multivariate Long-Short Term Networks	-0.044	0.044	0.120

Table 5 PM10 Performance Evaluation

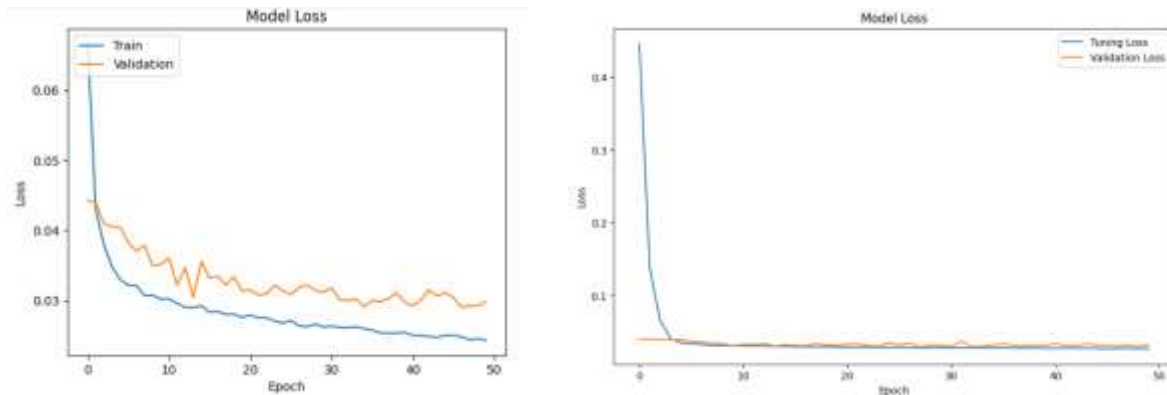
The predictive models produced challenging results. The Random Forest Regressor and Support Vector Regressor showed limited ability to predict PM10 variability, with R2 values of 0.064 and -0.020, respectively. The Multivariate Long-Short Term Memory Networks also had a negative R2 value of -0.044, indicating a lack of precise predictive ability. Overall, these results do not align with the project's goals of forecasting urban emissions.

PM2.5:

Model	R ²	MSE	MedAE
Random Forest Regressor	0.064	0.586	0.597
Support Vector Regressor	0.072	0.842	0.499
Multivariate Long-Short Term Networks	0.191	0.035	0.121

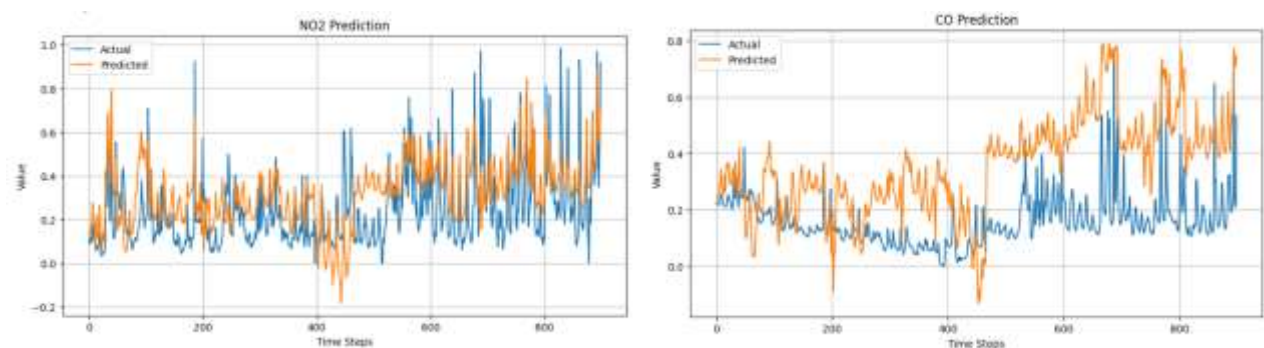
The Random Forest Regressor and Support Vector Regressor models have low R2 values, indicating inadequate prediction accuracy. Despite showing a slight improvement, the Multivariate Long-Short Term Memory Networks model offers the highest R2 value and lower prediction errors, making it the most reliable among the three models.

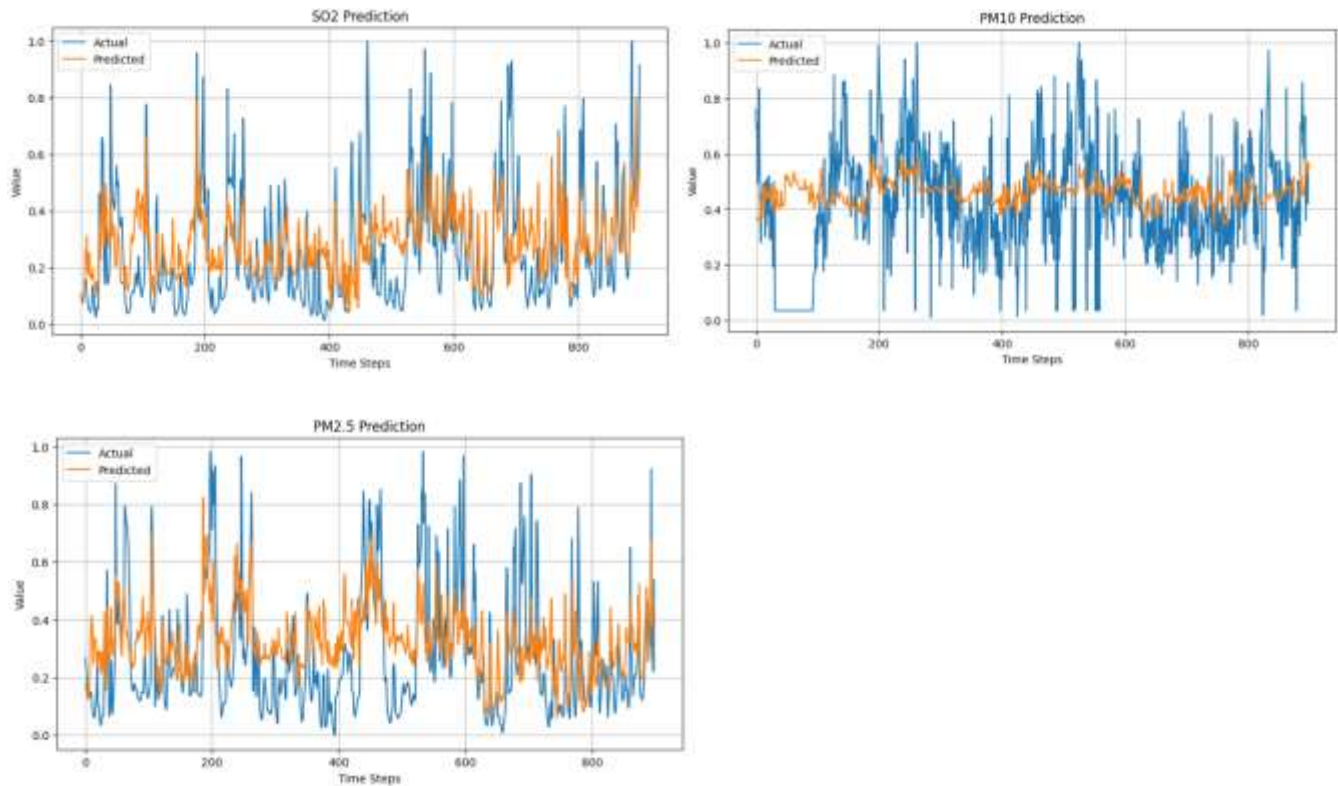
Figure below shows the Multivariate LSTM loss on training and tuning.



Compared to the base model, the model loss after tuning reveals how quickly the tuning loss (blue) decreases from 0.3 to 0.05 and then levels off, while the the validation loss(orange line) decreases slowly and levels off closely to the training loss line. This suggests that the model is converging to a point where additional training does not result in significant overfitting. The model appears to be performing well, with both the training and validation loss stabilizing to a low value, indicating good generalization up until the 50th epoch.

The Figures below are the actual vs predicted values using the Multivariate LSTM model.





4.1 Discussion:

This study conducted a comprehensive analysis to understand the impact of vehicular fuel types on air quality under different meteorological conditions. The project utilized exploratory data analysis (EDA) and predictive modeling to provide insights into the issue.

The EDA showed that diesel vehicles, particularly heavy goods vehicles (HGVs), have a significant positive correlation with NO₂ and particulate matter (PM_{2.5} and PM₁₀), indicating that they are major contributors to urban air pollution. Also, petrol cars have a correlation with PM₁₀, but to a lesser extent. Meteorological conditions have a significant impact on pollutant dispersion, with wind speed showing a consistent negative correlation with all pollutants. Temperature and humidity have a more complex relationship with pollutants, suggesting variable impacts on chemical reaction rates and pollutant deposition processes.

Furthermore, Granger's causality tests revealed a predictive relationship between diesel HGV emissions and SO₂ levels, which was not initially apparent from the correlation analysis. This indicates that past emissions from these vehicles can be used to forecast future SO₂ concentrations.

Regarding predictive modeling, the analysis of NO₂ levels showed that Multivariate Long-Short Term Memory Networks (LSTMs) outperformed Random Forest and Support Vector Regressor (SVR) models with lower error metrics. However, it explained less variance according to the R² statistic. For SO₂, the models struggled, with LSTMs producing a negative R², indicating a poor fit to the data. This suggests that the current model configurations may not capture the complexities of SO₂ emissions, or the data may not contain sufficient information for accurate SO₂ prediction.

The LSTM showed promise for predicting CO, with a reasonable R² and low error metrics, indicating the potential for developing effective prediction tools. However, the prediction of particulate matter (PM₁₀ and PM_{2.5}) was challenging for all models, with generally low R² values and high error rates. The LSTMs did offer the lowest errors for PM_{2.5} prediction, providing a basis for further model refinement.

The project's findings highlight the significant impact of diesel vehicle emissions on urban air quality and the crucial role of weather conditions in modulating pollutant concentrations. While predictive models have shown potential, particularly the LSTM for certain pollutants, further development is necessary to improve accuracy and reliability.

Conclusion

To better understand the complex relationship between vehicle fuel types and air quality in urban environments, set against the backdrop of varying meteorological conditions, this dissertation undertook an analytical journey. The research is significant because it addresses the growing concerns about urban air pollution, which has a significant impact on public health, environmental sustainability, and global climate change.

Urban areas, with their dense populations and high traffic volumes, present a unique challenge in understanding and managing air quality. The combustion of various fuel types in vehicles releases a range of pollutants, each contributing differently to the air quality depending on their chemical composition and the prevailing meteorological conditions [15]. The study's relevance is further heightened by the evolving landscape of vehicular technology and fuel usage. With the advent of more stringent environmental regulations and the push towards sustainable transportation, understanding the impact of different fuel types on urban air quality has never been more crucial [29].

This study, therefore, not only confronts the gap in the existing literature but also provides valuable insights for policymakers and urban planners to formulate strategies to minimize air pollution. The investigation was structured around three core research questions that explored different aspects of the relationship between vehicular fuel type emissions, urban air quality and varying meteorological conditions. The first question delved into the correlation between different vehicular fuel type emissions and the levels of key pollutants in urban air quality. The second question explored how meteorological conditions influence the impact of emissions from various fuel types on urban air quality. Finally, the third question examined the potential of machine learning models in predicting air pollution levels based on the distribution of vehicle fuel types and meteorological data. These questions were pivotal in guiding the research and highlighting the significance of the findings within the broader context of environmental science and urban planning.

The study unveiled that diesel vehicles, particularly heavy goods vehicles (HGVs), are significantly correlated with higher levels of NO₂ and particulate matter (PM_{2.5} and PM₁₀). This finding underscores the substantial role of diesel emissions in urban air pollution. Petrol cars also showed a correlation with PM₁₀, albeit to a lesser extent. The role of meteorological conditions emerged as a critical factor, with wind speed consistently influencing pollutant dispersion. The complex interplay of temperature and humidity was also observed, affecting chemical reactions and deposition processes. A notable discovery was made through Granger's causality tests, which highlighted a predictive relationship between diesel HGV emissions and SO₂ levels. In the realm

of predictive modeling, Multivariate Long-Short Term Memory Networks (LSTMs) demonstrated potential in forecasting CO and NO₂ levels. However, the study encountered challenges with SO₂ and particulate matter predictions, indicating the need for further refinement of these models. The findings from this research not only enhance our understanding of urban emissions but also lay the groundwork for future research and policy development aimed at improving urban air quality. The dissertation makes a significant contribution to the field by providing a detailed analysis of the impact of diesel vehicle emissions on urban air quality, a nuanced understanding of the role of meteorological conditions in pollutant dispersion and concentration, and advancements in the application of LSTMs for air quality prediction. The findings of this study address the research questions directly and establish a clear correlation between diesel fuel and key urban pollutants. The study also details the influence of weather on emissions and highlights the potential and limitations of machine learning models in pollution prediction. Therefore, this study stands as evidence of the complex interplay of emissions from different vehicular fuel types, meteorological conditions, and predictive modeling in understanding and mitigating urban air pollution.

5.1 Limitations and Future Work

The journey of this research, while enlightening, was not without its limitations. A notable constraint was the unavailability of data which also led to several others. The primary data, Traffic data, proved challenging to come about, as available sources either lacked temporal granularity or the variables required for analysis. Due to this, the study depended on series of feature engineering to ensure a robust analysis.

The traffic data had some large gaps, which were filled in by interpolation. However, this method of estimation could have resulted in less accurate data. Additionally, the expected proportions of each fuel type within the traffic counts were calculated using statistical data for each vehicle category, which only provide estimates and not actual measures. To estimate emissions from each vehicle fuel type proportion, the study relied on the use of extrapolated emission factors for the year 2022. The study also made some assumptions, such as generalizing emissions for heavy goods vehicles (HGVs) and the difficult exclusion of hybrid cars due to the unavailability of data, which might have provided additional insights into the emissions landscape.

All these were necessary compromises, given the data constraints. These limitations, while acknowledged, underscore the challenges faced in environmental research and the importance of continuous data improvement. There is a vast field of research for future inquiry. The inclusion of additional vehicle features (such as fuel type, car model, engine size, etc.) to the data set, as it becomes available, would certainly enrich our understanding of emissions profiles. The inclusion of direct NO_x air quality data would allow for a more comprehensive analysis of nitrogen oxide emissions. A separate analysis of rigid and articulated HGVs and Hybrid vehicles could offer a more detailed picture of their respective contributions to urban air pollution. In terms of methodology, future research could benefit from improved data collection methods and the exploration of more sophisticated machine learning algorithms. Such advances could significantly enhance the precision and depth of research in this field. Predictive modeling, particularly employing advanced machine learning and Time series algorithms, holds considerable promise for future exploration. In conclusion, this study has established a foundation for understanding the intricate dynamics of urban emissions. It emphasizes the significance of addressing current limitations and indicates exciting possibilities for future research. The pursuit of knowledge in urban emissions is far from complete, and this research serves as a stepping stone towards a more profound comprehension and more effective management of urban air quality.

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APPENDIX

TABLE 1: TRAFFIC DATA

	Motorcycles	Cars	Buses_and_coaches	Light_goods_vehicles	Heavy_goods_vehicles
count	39324.000000	39324.000000	39324.000000	39324.000000	39324.000000
mean	59.152121	2018.885286	29.359094	488.329595	114.443470
std	59.135469	1834.897401	26.389881	460.810773	124.901692
min	0.000000	7.000000	0.000000	0.000000	0.000000
25%	15.000000	859.000000	11.000000	175.000000	27.000000
50%	40.000000	1482.000000	22.000000	314.000000	61.000000
75%	95.000000	2718.000000	45.000000	758.000000	182.000000
max	449.000000	13732.000000	189.000000	4311.000000	1235.000000

TABLE 2: AIR QUALITY DATA

	AQI	CO	SO2	PM2.5	NO2	PM10
count	17305.000000	17305.000000	17305.000000	17305.000000	17305.000000	17305.000000
mean	1.456227	251.727561	8.293220	6.854734	17.896976	14.886645
std	0.768481	84.760792	7.673756	8.305755	14.825756	9.015142
min	1.000000	146.870000	1.300000	0.500000	0.000000	-5.300000
25%	1.000000	198.600000	4.170000	2.260000	7.618820	9.300000
50%	1.000000	236.990000	6.320000	4.000000	12.987850	13.400000
75%	2.000000	277.040000	9.780000	8.080000	23.473500	18.600000
max	5.000000	1535.420000	188.830000	115.890000	119.011960	171.200000

TABLE 3: WEATHER DATA

	temperature	pressure	humidity	wind_speed	wind_dir	rain_vol	snow_vol
count	43824.000000	43824.000000	43824.000000	43824.000000	43824.000000	43824.000000	43824.000000
mean	11.826807	1012.813162	77.377236	3.886265	195.055198	0.090107	0.001114
std	6.349158	11.953370	15.833298	2.287779	97.792782	0.465331	0.025385
min	-5.600000	962.000000	17.000000	0.000000	0.000000	0.000000	0.000000
25%	7.270000	1005.000000	68.000000	2.100000	110.000000	0.000000	0.000000
50%	11.300000	1014.000000	81.000000	3.600000	220.000000	0.000000	0.000000
75%	16.180000	1021.000000	90.000000	5.140000	270.000000	0.000000	0.000000
max	38.590000	1049.000000	100.000000	18.330000	360.000000	42.160000	1.270000